Marginal, not per-scan: a pre-registered five-cohort audit of conformal lesion-wise false-discovery control for multiple sclerosis lesion segmentation across sites and slice thicknesses

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Manuscript draft 1.0, 2026-10-31. Research software, not the medical device. Every number is traced in NUMBERS_CHECK.md.

Abstract

Background. Conformal selection can wrap any lesion segmenter with a distribution-free "precision dial": among the candidate lesions of a scan, report only those whose conformal p-values pass Benjamini–Hochberg, so that the expected share of false lesions (the lesion-wise false-discovery rate, FDR) is at most α. The promise rests on exchangeability between the calibration data and the scan at hand. Multiple sclerosis (MS) imaging routinely breaks that assumption. Sites differ, much FLAIR is still acquired as 2D slices of 4–8 mm, and public cohorts often sit inside the segmenter's training data.

Purpose. To measure, under a pre-registered protocol, where lesion-wise conformal false-discovery control holds and where it fails for a state-of-the-art MS segmenter: across sites, from thin to thick slices, with and without fold ensembling, and on healthy brains.

Methods. We used five public patient cohorts (304 scans, 245 patients) and 100 healthy controls, segmented with the five folds of FLAMeS. Two patient cohorts are delivered on a 1-mm grid: 10 scans are natively thin, 110 are 2–5 mm acquisitions resampled to 1 mm and 5 have unknown native thickness. Three cohorts are natively 2D, with 3.9–8.3 mm slices. We excluded every scan in FLAMeS's training data. We resampled and split by patient, never by scan. The protocol was committed before any outcome was computed (2026-10-21; git commit 9951494; SHA-256 1b5b39a8…6016). A separately pre-registered addendum covers the controls (2026-10-24; commit caa47e7). We compared conformal BH (P1, marginal FDR) with a scan-level exceedance threshold (P2, "CALM-FDX": $P(FDP > \gamma) \leq \delta = 0.1$). P2 is an application of conformal risk control and Learn-then-Test, not new theory. We also compared risk-control, Learn-then-Test and precision-matched baselines, plus two scan-aware baselines analysed post hoc.

Results. No violation of marginal FDR control was detected in the 16 pre-registered held-out-site comparisons (14 on sites FLAMeS never trained on). This is weaker than validity. One thick-slice site was undecided at α = 0.1 (Muslim: realised FDR 0.122 [95 % CI 0.062, 0.197]). Three comparisons passed only because almost nothing was selected. With 1-mm calibration, realised FDR on the two large thick-slice sites was 2.6–3.2 times its within-site value. Post hoc, on the third of thick-slice scans with many false candidates, it was 0.196 [0.120, 0.282]. The per-scan guarantee failed on thick slices calibrated on thin ones: 21 % [13, 30] (MS3SEG) and 29 % [17, 44] (Muslim) of scans exceeded γ = 0.1, against a 10 % budget. Only 10 or more local calibration patients restored it, and then only by near-abstention (3.85 % and 0.65 % of lesions found). Thickness effects were confounded with site. Averaging the five folds raised candidate AUROC from 0.67–0.76 to 0.78–0.82. It doubled lesion sensitivity at α = 0.2 (0.102 → 0.203, leave-one-site-out) and lowered pooled realised FDR. On one site, however, it raised realised FDR at α = 0.1 (0.052 → 0.122). On healthy brains, 1-mm-calibrated selection flagged 14/100 (α = 0.1) and 24/100 (α = 0.2) controls; both results were undecided and above nominal. Without selection, the segmentation marked a lesion in 85/100 healthy brains and met a brain dissemination-in-space rule in 19/100 [11.8, 28.1 %].

Conclusions. Marginal lesion-wise FDR control survived site and thickness shift in the sense that no violation was detected. Per-scan control did not survive. Lesion-wise conformal guarantees for MS should be calibrated with the deployed segmenter on the target site or acquisition type. They leave little power, and they do not cover healthy or screening populations. We translated these findings into a slice-thickness reliability card that is off by default and shows no false-lesion rate. This is translation, not clinical validation.

Keywords: conformal selection; false discovery rate; false-discovery exceedance; Learn-then-Test; multiple sclerosis; lesion segmentation; distribution shift; slice thickness; pre-registration; training-set contamination.

1 Introduction

Automated segmentation of MS white-matter lesions on FLAIR has reached voxel Dice of about 0.7 on external data with nnU-Net-class models (Isensee et al., 2021; Dereskewicz et al., 2025; Wiltgen et al., 2024). A reader of a single scan, however, needs a different number: how many of the lesions the software marks are false. Diagnostic rules make each lesion count. The McDonald criteria require lesions in several characteristic regions to establish dissemination in space (Thompson et al., 2018; Montalban et al., 2025). A few false periventricular or juxtacortical marks can turn a negative scan positive.

Conformal inference offers a principled answer. Conformal p-values computed against a calibration set of known false candidates are valid and positively dependent (Bates et al., 2023). Benjamini–Hochberg (BH) selection on them controls the false-discovery rate of the reported set (Benjamini and Hochberg, 1995; Jin and Candès, 2023). The resulting "precision dial" needs no model assumptions. Its one proviso is exchangeability between the calibration candidates and the false candidates of the new scan. Risk-control and Learn-then-Test (LTT) frameworks provide analogous guarantees with the scan as the exchangeable unit (Angelopoulos et al., 2024; Angelopoulos et al., 2025; Bates et al., 2021).

MS imaging stresses this proviso in four ways. First, scanners and protocols differ between sites. Second, 2D FLAIR with 4–8 mm slices is still common in routine care, although consensus guidelines recommend 3D FLAIR (Wattjes et al., 2021); thick slices change both lesion appearance and the segmenter's errors. Third, false candidates within one scan share that scan's quality and anatomy, so they are not exchangeable as single units.

Fourth, the public MS cohorts used for evaluation are often part of the segmenter's training data, which turns an external test into an in-distribution one (Kapoor and Narayanan, 2023; Varoquaux and Cheplygina, 2022).

This paper does not propose a new conformal method. It is a pre-registered, reproducible audit of lesion-wise conformal false-discovery control on real MS data, with the safeguards such a claim needs: a forensic check of which scans the segmenter trained on, the patient as the unit of splitting and resampling, held-out sites, multiplicity-aware intervals, healthy controls and plainly reported negative results. Our earlier internal drafts made claims that this audit retracts; we list them (Appendix B).

Contributions.

1. A pre-registered five-cohort evaluation of lesion-wise conformal FDR (P1) and per-scan false-discovery exceedance (FDX, P2) control, with leave-one-site-out and thin-to-thick-slice transport designs, in which every scan in the segmenter's training data is excluded (§6.2–6.3).

2. A comparison with pre-registered (conformal risk control, LTT, precision-matched) and post-hoc scan-aware baselines (one candidate per patient; hierarchical conformal e-values) (§6.4).

3. A slice-thickness reliability table: validity, power and lesion-size-resolved detection per thickness band. We state its confounding with site plainly (§6.6).

4. Evidence that fold ensembling doubles power at the guarantee without fixing validity (§6.7).

5. A recalibration-size analysis: how many local patients restore validity, and at what power (§6.8).

6. A pre-registered healthy-control analysis, which replaces an unfounded earlier claim about dissemination in space on healthy brains (§6.9).

7. A description of how the findings were translated into a conservative, default-off software feature, explicitly not a clinical validation (§7).

2 Related work

Conformal selection and FDR. Conformal p-values against a shared calibration set satisfy PRDS, so BH on them controls FDR at $\alpha \cdot \pi_0 \leq \alpha$, where π_0 is the share of nulls among the tested units (Bates et al., 2023). Jin and Candès (2023) cast prediction-based selection in this framework, and Marandon et al. (2024) extended it to adaptive scores. Under covariate shift, weighted conformal p-values restore validity when the likelihood ratio is known or can be estimated (Tibshirani et al., 2019; Jin and Candès, 2026). More general deviations from exchangeability cost a quantifiable coverage gap (Barber et al., 2023). Mondrian (group-conditional) calibration calibrates within strata such as sites or thickness bands (Vovk et al., 2005).

Grouped data and scan-level risk control. When observations are clustered, exchangeability holds at the cluster level at best. Dunn et al. (2023) calibrate by subsampling one observation per group. Lee and Ren (2025) construct hierarchical and subsampling conformal e-values with e-BH (Wang and Ramdas, 2022); their guarantee assumes group sizes independent of the observations. Conformal risk control (Angelopoulos et al., 2024) and LTT (Angelopoulos et al., 2025) control the expectation, or a high-probability bound, of any bounded loss over exchangeable units. With the scan as unit and the per-scan false-discovery proportion (FDP), or the indicator

{FDP > γ}, as loss, they control scan-level FDR or FDX for a global threshold. Non-exchangeable risk control trades the guarantee for weights over the calibration data (Farinhas et al., 2024).

Simultaneous and exceedance bounds. FDX control dates back to Genovese and Wasserman (2006). Reference-family bounds (Blanchard et al., 2020), the exact joint law of conformal p-values (Gazin et al., 2024) and everywhere-valid FDP bounds for conformal selection (Song et al., 2026) give post hoc or simultaneous control. All of them assume exchangeability of the individual candidates.

Conformal methods in medical image segmentation. Recent work controls voxel-level miss rates for 3D lesions (Tan et al., 2025), image-level false-positive proportions in segmentation masks (Mossina and Friedrich, 2025) and per-scan nodule miss rates by risk control (Hulsman et al., 2025). Lambert et al. (2024) estimate volumes under covariate shift. Adhikary et al. (2026) showed that per-organ recall risk control in CT fails to transfer under clinical domain shift. In MS, conformal prediction has been applied to disease course from registry data (Sreenivasan et al., 2025). MS lesion uncertainty has otherwise been evaluated without distribution-free guarantees (Molchanova et al., 2025; Malinin et al., 2022).

Position of this work. We found no study of lesion-wise conformal FDR or FDX control for MS segmentation that tests transport across sites and slice thickness while controlling for the segmenter's training-set overlap. We make no stronger novelty claim. The methods are standard, and the per-scan procedure is an application of risk control and LTT (§3.2). Before submission, the literature search must be refreshed (Appendix D, owner checklist).

3 Problem setting and procedures

3.1 Candidates and lesion-wise error

For a scan, the segmenter outputs a voxelwise lesion probability map. A *candidate* is an 18-connected component of {p ≥ 0.5} whose physical volume is at least 3 mm³, computed with the scan's own voxel volume. Its *score* s is the mean probability over the component; higher scores are more lesion-like. *Expert lesions* are 18-connected components of the expert mask of at least 3 mm³ ("detectable"). *Clinical* lesions are those of at least 14.14 mm³ (a sphere of 3 mm diameter). Candidates are matched one-to-one to expert lesions by maximum-IoU Hungarian assignment (Kuhn, 1955). The pre-registered primary matching is lenient (IoU > 0); a strict variant (IoU ≥ 0.10) is a sensitivity analysis. A candidate is *true* if it is matched; unmatched fragments are *false*.

A selection rule reports a subset R of a scan's candidates. Its false-discovery proportion is FDP = |R ∩ F| / max(|R|, 1), where F is the set of false candidates. We report:

- the **realised FDR**: the mean per-scan FDP over test scans, with FDP = 0 when nothing is selected;
- the **realised FDX**: the share of test scans with FDP > γ;
- the **end-to-end sensitivity**: selected-and-matched expert lesions divided by all detectable expert lesions.

On a lesion-free scan, FDP is the indicator of at least one reported lesion. Each promise then becomes a bound on the healthy false-alarm rate.

3.2 Procedures

P1 — conformal BH (primary FDR procedure). The null set N contains the scores of all false candidates of all scans of the calibration patients, with size n. For a test candidate with score s, p = (1 + #{u ∈ N : u ≥ s}) / (n + 1).

BH at level α is applied within each test scan. If the test scan's false-candidate scores are exchangeable with N, these p-values are valid and PRDS, and E[FDP] ≤ α·π0 ≤ α (Bates et al., 2023; Jin and Candès, 2023). Candidate-level exchangeability is exactly what a scan-level random effect breaks. We measured that effect as the intraclass correlation of false-candidate scores within scans.

P2 — CALM-FDX, single-threshold form (primary FDX procedure). For each calibration patient (first timepoint only) and each threshold λ on the grid {0, 0.001, …, 1, ∞}, FDP_i(λ) is the FDP of {s ≥ λ}. The scan's *safe threshold* t_i is the smallest grid value λ such that FDP_i(λ′) ≤ γ for every grid λ′ ≥ λ. With K calibration patients, $\hat{\lambda}$ is the $\lceil (K+1)(1-\delta) \rceil$-th smallest t_i. If that index exceeds K, $\hat{\lambda} = \infty$ and the procedure abstains. Because FDP_test($\hat{\lambda}$) > γ implies t_test > $\hat{\lambda}$, and t_test is exchangeable with t_1, …, t_K when patients are exchangeable, P(FDP_test > γ) ≤ δ. This is the split-conformal quantile argument (Vovk et al., 2005). Equivalently, it is conformal risk control with the binary loss 1{t_i > λ} (Angelopoulos et al., 2024). Its high-probability (PAC) analogue is LTT (Angelopoulos et al., 2025). We call it CALM-FDX to keep continuity with our earlier drafts, but **it is an application of these frameworks, not new theory**. The earlier two-layer variant with per-scan region certificates was not re-run (§9).

Baselines (pre-registered). *B1, precision-matched threshold*: the lowest threshold whose pooled calibration micro-FDR is at most α; it carries no guarantee. *B2, conformal risk control*: a global threshold with the monotonised per-scan FDP as loss, which gives E[FDP_test] ≤ α under scan exchangeability (Angelopoulos et al., 2024). *B3, LTT*: Hoeffding–Bentkus p-values with fixed-sequence testing over thresholds and per-scan FDP as loss; with probability at least 0.9 over the calibration draw, E[FDP_test] ≤ α (Angelopoulos et al., 2025). B2 and B3 use one scan per calibration patient.

Baselines added post hoc (§5.5). *SUB*: one false candidate drawn at random per calibration patient, conformal p-values against this subsampled null and BH within the scan (Dunn et al., 2023). Each p-value is marginally valid under patient exchangeability, but BH's PRDS argument is not re-established, so its FDR control is empirical. The procedure is randomised (200 draws). *LR-hier*: hierarchical conformal e-values with weights 1/N_k and e-BH over the whole batch of test scans of the held-out site (Lee and Ren, 2025; Wang and Ramdas, 2022). Its guarantee is a batch FDR and assumes group sizes independent of the observations, which is doubtful here: candidate counts track lesion load.

Table 3. The procedures and what each one guarantees.

procedure	exchangeable unit assumed	error controlled	type of guarantee	randomised	status
P1 conformal BH	false candidates (pooled)	per-scan FDR ≤ α	marginal (over calibration and test)	no	pre-registered, primary
P2 CALM-FDX threshold	patients (one scan each)	P(FDP_scan > γ) ≤ δ	marginal	no	pre-registered, primary
B1 precision-matched	—	none	—	no	pre-registered baseline
B2 conformal risk control	patients	per-scan FDR ≤ α	marginal	no	pre-registered baseline
B3 Learn-then-Test	patients	per-scan FDR ≤ α	PAC, δ = 0.1	no	pre-registered baseline
SUB one candidate per patient + BH	patients	none proven (each p-value valid)	—	yes	post hoc baseline
LR-hier hierarchical e-values + e-BH	patients, sizes independent of data	batch FDR ≤ α	marginal	no	post hoc baseline

4 Data

4.1 Cohorts

Table 1 lists the cohorts. The five patient cohorts are MSLesSeg (Rondinella et al., 2025; MSLesSeg dataset, 2025), the Ljubljana 3D MR MS dataset (Lesjak et al., 2018), MS3SEG (Bawil et al., 2026; MS3SEG dataset, 2026), the Baghdad cohort of Muslim et al. (2022) (Brain MRI dataset of multiple sclerosis with consensus manual lesion segmentation and patient meta information (data), 2022) and PediMS (Popa et al., 2025; PediMS dataset, 2025). The healthy controls are the 100 controls of SibBMS (Tuchinov et al., 2025).

Two acquisition facts matter for everything that follows:

- **The "1-mm" cohorts are 1-mm grids, not 1-mm acquisitions.** MSLesSeg's release includes an acquisition table that lists native FLAIR slice thicknesses of 2–5 mm: 43 scans at 2–2.3 mm, 52 at 3 mm, 15 at 4–5 mm and 5 unknown. The images are registered and resampled to MNI 1 mm. Only the Ljubljana scans are natively thin (3D, 0.8 mm). Throughout, "1-mm sites" means sites *delivered* on a 1-mm grid.
- **The thick-slice cohorts are natively 2D**, with slice spacing of 3.9 mm (PediMS), 4.86–8.3 mm (Muslim) and 6.0–8.1 mm (MS3SEG). They were analysed on their native grids.

We skull-stripped MS3SEG and Muslim with HD-BET 2.0.1 (Isensee et al., 2019). PediMS is distributed skull-stripped. MSLesSeg and Ljubljana are used as distributed (MNI 1 mm).

The expert masks follow different protocols. MS3SEG's MS-lesion class excludes periventricular hyperintensities that its raters judged to be non-MS, so FLAMeS marks there count as false. Muslim masks come from a sequential correction chain, and Muslim's in-plane pixel size was not reliably available and was estimated (±5 % per axis). PediMS has one rater validated by two specialists and includes one patient with Baló's concentric sclerosis.

Table 1. Cohorts (5-fold ensemble arm; candidates and labels as in §3.1).

cohort	origin, scanner	scans / patients	native FLAIR geometry	grid analysed	detectable expert lesions	candidates (false)	candidate AUROC, 5-fold mean [95 % CI]	AUROC, single folds (range)
MSLesSeg	Italy (Catania); 8 scanner models, 1.5/3 T	115 / 75	2D/3D, native 2–5 mm (43 at 2–2.3, 52 at 3, 15 at 4–5, 5 unknown)	MNI 1 mm (delivered)	3344	3788 (1257)	0.818 [0.798, 0.837]	0.735–0.758
Ljubljana-10	Slovenia (Ljubljana); Siemens 3 T	10 / 10	3D, 0.47 × 0.47 × 0.8 mm	MNI 1 mm (delivered)	667	517 (82)	0.802 [0.729, 0.883]	0.673–0.701
MS3SEG	Iran; Toshiba 1.5 T	100 / 100	2D, slice spacing 6.0–8.1 mm (median 7.2)	native	3354	3402 (1162)	0.802 [0.786, 0.816]	0.699–0.720
Muslim	Iraq (Baghdad), ~20 centres; 1.5 T	51 / 51	2D, 4.86–8.3 mm (median 6.0); in-plane pixel size estimated	native	1819	1540 (568)	0.784 [0.741, 0.822]	0.731–0.746
PediMS	one centre; Siemens Skyra 3 T; 9 children	28 / 9	2D, 0.75 × 0.75 × 3.9 mm	native	1104	1110 (321)	0.801 [0.774, 0.864]	0.683–0.715
SibBMS healthy controls	Siberia (Russia), one site	100 / 100	not verified; images registered to an MNI-type 1-mm template	template 1 mm	0	263 (263); fold 0: 306	—	—

4.2 Training-set overlap (contamination)

FLAMeS was trained on 668 FLAIR scans from 11 sources (Dereskewicz et al., 2025). Two public MS cohorts are among them: 20 ISBI-2015 scans (Carass et al., 2017) and 20 Ljubljana scans (listed as PubMRI) (Lesjak et al., 2018). MSSEG-2016 scans (centres 01, 03, 07 and 08) are also included (Commowick et al., 2018). A segmenter evaluated on its own training scans is not being tested externally (Kapoor and Narayanan, 2023). We therefore handled overlap as follows (Table 2).

- **ISBI-2015** is not used anywhere in this paper. An analysis in our earlier draft that used it is withdrawn.
- **Ljubljana.** The released weights contain nnU-Net's dataset fingerprint (per-case spacing and brain bounding box of all 668 training cases). Twenty consecutive entries match Ljubljana patients 01–15 and 21–25 exactly, with zero voxels of disagreement on all three axes. The training logs show that each fold trains on four fifths of the cases. These 20 scans are excluded from every analysis, including calibration. The remaining 10 patients (16–20, 26–30; "Ljubljana-10") were never trained on by any fold. Their **site**, however, was in training. We therefore never count Ljubljana-10 as an unseen-site test. Its two leave-one-site-out cells are reported but flagged (†). A post-hoc analysis removes Ljubljana from every calibration pool (§6.5).
- **MSLesSeg, MS3SEG, Muslim, PediMS** are not among FLAMeS's training sources. MSLesSeg served as an external test set in the FLAMeS publication. For the three thick-slice cohorts, the exclusion rests on the dataset and model descriptions, not on fingerprint forensics.
- **SibBMS controls** are not among the training sources. The fingerprint has 668 entries, none compatible with the SibBMS template grid or any control's brain box, and SibBMS was released after FLAMeS was submitted.

Table 2. Segmenter training-set overlap and licences.

cohort	in FLAMeS training?	used here as	in LST-AI training?	licence (data)
MSLesSeg	no (FLAMeS external test)	calibration and test	no	CC BY 4.0 (the article itself is non-commercial; we use the data only)
Ljubljana (open_ms_data / 3D MR MS)	**20 of 30 scans** (patients 01–15, 21–25)	only the 10 never-trained scans; site flagged as seen	no (LST-AI test set)	**CC BY-NC-ND per the laboratory's page** (a download mirror states CC BY); we apply the stricter licence: academic use only
ISBI-2015	**yes (20 scans)**	**not used**	—	challenge terms
MS3SEG	no known overlap	test, and calibration in pooled designs	not used	CC BY 4.0 (data); code MIT
Muslim 2022	no known overlap	as MS3SEG	not used	CC BY 4.0
PediMS	no known overlap	as MS3SEG	not used	CC BY 4.0
SibBMS healthy controls	no (fingerprint, release dates)	test only (controls)	not run (pending)	stated as both CC BY 3.0 and CC BY-NC-SA 3.0; we apply the stricter: research use only

The FLAMeS weights (v2.0) are CC BY 4.0 (Dereskewicz et al., 2025). LST-AI's code is MIT-licensed. The HD-BET weights are licensed for non-commercial use.

5 Study design and statistics

5.1 Pre-registration and its limits

The analysis protocol (`docs/research/paper/PROTOCOL-v0.3.md`) was committed on 2026-10-21 at 04:21 CEST (commit 9951494). At that point, only the input inventory had been computed: file hashes, grids and value ranges. No candidate label, selection, FDP, recall or AUROC had been computed on the new data. The SHA-256 of the protocol file at that commit is 1b5b39a8e5cc3c3af6cfce22d64c16832e0ccc71dbaf5ba35b11db15b93c6016 (committed blob, LF line endings). Implementation clarifications 1–4 (commit c0c5fa1, 04:35) and the gate deviation 5 (commit 50c9b08, 04:41) were logged before any outcome. The first results were committed at 04:59 (8a420bc).

The healthy-control addendum (`PROTOCOL-v0.3-addendum-controls.md`) was committed on 2026-10-24 at 15:52 CEST (commit caa47e7; SHA-256 bdae74ec6373daa0685c49cc4f57fed73445716b92d469d9658cede764f87213). It was written before any control outcome on the ensemble maps or on the 70 never-analysed fold-0 control maps.

Limit, stated plainly. The pre-registration lives in the project's version-controlled repository, not in a public registry (Nosek et al., 2018). For the main protocol, the external timestamp is weak: the branch reached the public remote (GitHub) only at 05:09 CEST on 2026-10-21, together with the first results. Its precedence over the results therefore rests on the git history, whose author timestamps cannot be verified independently. The addendum is stronger: it was pushed to the public remote at 13:52:31 UTC on 2026-10-24, 71 minutes before its results (15:03:30 UTC). These times come from GitHub push events. A three-scan processing pilot on one scan per thick-slice cohort (FLAMeS fold 0, 2026-10-18) preceded the protocol. Its outcomes were not inspected, and it is disclosed in protocol entry 7.

5.2 Segmenter and arms

FLAMeS (Dereskewicz et al., 2025) is a 3D full-resolution nnU-Net (Isensee et al., 2021) trained on FLAIR only. We ran each of its five folds separately (checkpoint_final, no test-time augmentation) and stored probability maps as uint8 with a scale factor; the maximum decoding error is about 0.002.

- **Ensemble arm (primary).** The voxelwise mean of the five fold maps, which is nnU-Net's own ensembling (Lakshminarayanan et al., 2017). Candidates are extracted from that map.
- **Fold-0 arm.** The pipeline of our earlier drafts. Folds 1–4 were run the same way to check whether fold 0 is typical.
- **Secondary score.** S_cons is the minimum over folds of the per-fold mean probability on an ensemble candidate; it is lesion-like only if all folds agree.
- **LST-AI** v1.2.0 (Wiltgen et al., 2024) was run on MSLesSeg and Ljubljana only, for an exploratory paired segmenter swap.

A reproduction gate (G1) compared the new fold-0 candidates with those of our earlier analysis: 125/125 scans had identical candidate counts and 4400/4400 candidates had identical one-to-one labels. One candidate's score differed by 0.0078, above the 0.003 tolerance. This is a documented boundary voxel, and the deviation was logged before any outcome (Table 11, entry 5).

5.3 Calibration designs

- **D1, within-site:** patient-grouped 5-fold cross-fitting within each site; leave-one-patient-out for Ljubljana-10 and PediMS.
- **D2, leave-one-site-out:** calibrate on all eligible patients of the other four sites, test on the held-out site.
- **D3, 1-mm grid → thick (primary transport design):** calibrate on MSLesSeg + Ljubljana-10, test on MS3SEG, Muslim and PediMS separately. D3r is the reverse direction.
- **D4, pooled:** patient-grouped 5-fold cross-fitting over all five sites.
- **D5, recalibration curve:** draw k patients of the target site ($k \in \{1, 2, 3, 5, 10, 20, 40\}$, 200 draws each), calibrate either on them alone (*site-only*) or on the 1-mm pool plus them (*augmented*), and test on the rest of the site.

Repeated timepoints of one patient (MSLesSeg, PediMS) are never split between calibration and test.

5.4 Outcomes, classification and multiplicity

The **32 primary cells** cover the held-out sites in D3 (3 thick-slice sites) and D2 (5 sites), for P1 at $\alpha \in \{0.1, 0.2\}$ (realised FDR) and P2 at $\gamma \in \{0.1, 0.2\}$ with $\delta = 0.1$ (realised FDX). All are for the ensemble arm with the primary score and lenient matching.

Intervals are 95 % patient-cluster percentile bootstraps with 2000 resamples (Davison and Hinkley, 1997). For FDX, the interval is the union with the exact Clopper–Pearson interval over scans (Clopper and Pearson, 1934). Each cell is classified as follows:

- *violated* if the lower bound exceeds the nominal level;
- *holds* if the upper bound is at most nominal + 0.05;
- *inconclusive* otherwise.

A non-violated cell with fewer than 10 selections or end-to-end sensitivity below 0.05 is flagged as *abstention*. "Holds" therefore means *no violation detected and the upper bound is within the margin*. It is not a proof of validity, and a "holds" cell can have a point estimate above α.

Under exact validity we would expect about 0.8 spurious one-sided "violations" among 32 cells. We therefore also report Bonferroni intervals (level $1 - 0.05/32$, 20 000 resamples) and a two-level bootstrap that resamples calibration and test patients (500 resamples), which gives the marginal view.

The pre-registered decision rules (protocol §10) were:

- Q1 (thin-to-thick transport) is *supported* if no D3 cell is violated and the MS3SEG and Muslim cells hold, and *refuted* if any D3 cell is violated.
- Q3 (ensemble) "helps" if the paired difference in sensitivity excludes 0 in its favour in D2 and D3 without a new violated cell.
- Q4 defines k* as the smallest k at which the mean realised error is at most nominal and the mean power is at least 90 % of within-site power.

For **healthy controls**, the primary cells are the ensemble arm with 1-mm-site calibration, P1 at $\alpha \in \{0.1, 0.2\}$ and P2. We report the share of the 100 controls with at least one reported lesion, with exact intervals and the same

classification rule. A secondary outcome applies the software's own brain dissemination-in-space (DIS) rule to the raw and selected lesions.

5.5 Post hoc analyses

Analyses written after results had been seen are labelled **[post hoc]** wherever they are cited. Three groups exist:

- explanatory checks of v0.3 (`v03_posthoc.py`), including the low-lesion-load analysis added after an independent re-implementation;
- a stratification of MSLesSeg by native thickness (`v03_mslesseg_native.py`);
- two analyses added at the manuscript stage (`v03m_manuscript_posthoc.py`): Ljubljana removed from every calibration pool, and the SUB and LR-hier baselines.

The manuscript-stage analyses were logged as protocol entry 10 and pushed to the public remote (commit 70b0831) immediately before they were run. The evidence for that order is the local git log of the push; no external timestamp was available for this push when we wrote this paper.

5.6 Deviations from the pre-registered protocol

Table 11. Deviations, clarifications and disclosures. Entries 1–10 refer to PROTOCOL-v0.3 §13; entries A1–A2 refer to the addendum §11.

#	when (relative to outcomes)	what	effect on reported numbers
1	before	FDX intervals: union of the patient-cluster bootstrap and the exact Clopper–Pearson interval (the percentile bootstrap degenerates at 0/n)	wider FDX intervals
2	before	Bonferroni intervals use 20 000 bootstrap resamples	none
3	before	Muslim ±10 % volume sensitivity rescales expert-lesion volumes only	none on primary
4	before	Gate G1 tolerance restricted to the 5 scans where a voxel crosses 0.5	none
5	before	G1 strict criterion failed on one candidate (score difference 0.0078 > 0.003; boundary voxel); analysis continued	none
6	after	fold-0 primary-cell classification and the Q3/Q4 verdict blocks were added to the script after its first run; the re-run reproduced every printed number; explanatory analyses labelled post hoc	none
7	disclosure	a three-scan thick-slice processing pilot (2026-10-18) preceded the protocol; outcomes not inspected	none
8	after	independent re-implementation of the patient analysis; wording corrected ("no violation detected", per-scan failure intervals, site confounding, mixed ensemble effect, near-abstention); post-hoc low-lesion-load analysis added	wording only; no pre-registered number changed
9	after	MSLesSeg is natively 2–5 mm (dataset acquisition table), not "unpublished geometry"; band B1 relabelled "≤ 1.5 mm as delivered"; post-hoc native-thickness stratification	labels only
10	before running, after all other	manuscript stage: Ljubljana	new post-hoc numbers (§6.4, §6.5)

	results	removed from every calibration pool; SUB and LR-hier baselines [post hoc]	
A1	after	control analysis: interrupted runs and two coding errors (a rounding helper; a key name) fixed; the fourth run reproduced every printed number; a separate re-implementation matched all counts exactly	none
A2	after	controls [post hoc]: visual check of four top-scoring false candidates; their FLAIR intensity; Fisher test of the 30 previously analysed vs 70 new controls	post-hoc numbers only
—	reporting	the two leave-one-site-out Ljubljana cells are reported as "site seen in segmenter training" rather than counted as unseen-site evidence	none
—	reporting	the earlier text stated that conformal BH was "the only procedure with no violated 1-mm → thick cell"; Learn-then-Test (B3) had none either (§6.4)	correction
—	not done	the LST-AI arm on the controls (addendum E1) is pending	—

5.7 Software, compute and reproducibility

Candidate extraction (18 min on 7 workers), the analysis (6 min) and every post-hoc analysis ran on a local CPU, with deterministic seeds (20261021, 20261024 and 20261031). FLAMeS and LST-AI inference ran earlier on CPU virtual machines (§Funding). Every input file is hashed in the inventory records. The cached candidate set is identified by its SHA-256 (2096109e…a6f4ec) and is not redistributed. Its licence conditions (Ljubljana: no derivatives) forbid sharing; scripts regenerate it from the public sources. The metric functions have 33 unit tests. Appendix C maps every result to its script and record, and `NUMBERS_CHECK.md` maps every number in this paper to its stored value.

Two independent re-implementations were made:

- **Patient analysis (2026-10-21).** It reproduced the candidates (for example, 3402 MS3SEG candidates, 1162 false) and the primary rates to three decimals, and found no leakage between calibration and test patients. Its code was not archived in the repository (Appendix D).
- **Control counts (2026-10-24).** Written as `v03c_controls_check.py`, with its own labelling, p-values and BH. All counts matched exactly.

Both re-implementations were written in separate AI-agent sessions, not by a second human analyst (see AI-assistance statement).

6 Results

6.1 Score quality

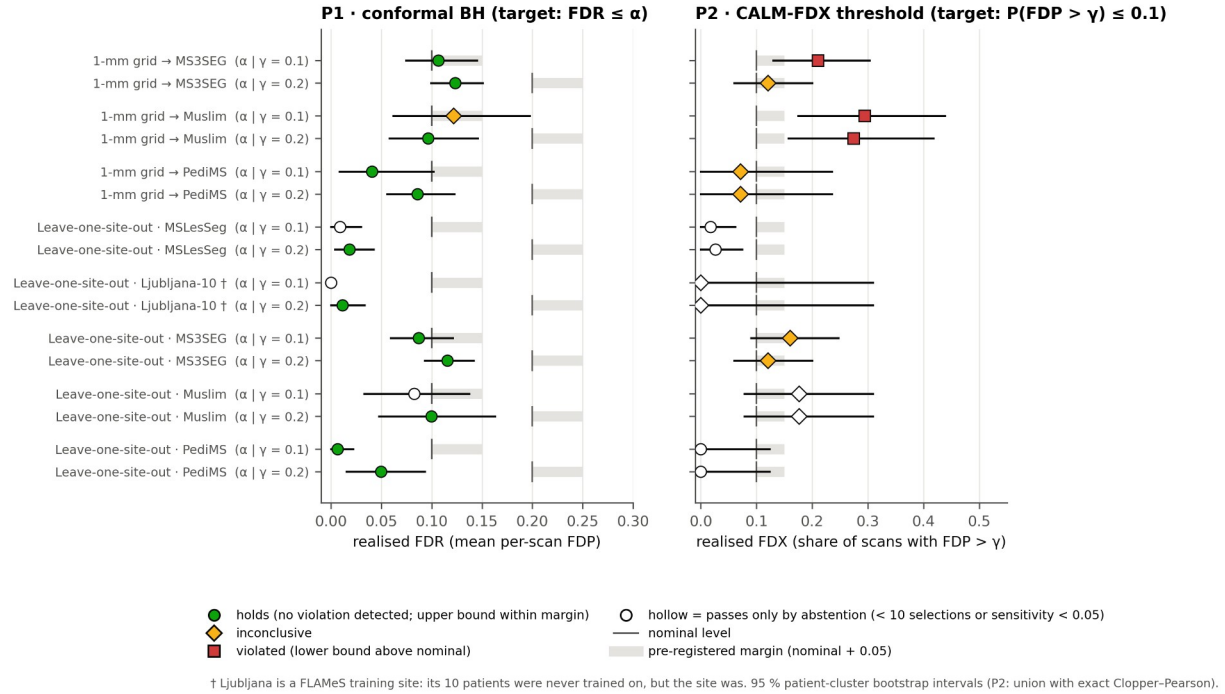
Every single fold ranked candidates with AUROC 0.67–0.76 (Table 1). The 5-fold mean reached 0.78–0.82 at every site. Candidate precision without selection was 0.63–0.84. The ensemble produced 8.2–11.6 false

candidates per scan. Fold disagreement alone (the negative standard deviation of the five per-fold candidate means) ranked candidates almost as well as the mean (AUROC 0.77–0.81).

6.2 Held-out sites: marginal FDR control (P1)

Table 4. The 32 pre-registered primary cells (ensemble arm). Entries: realised value [95 % CI] → class; end-to-end sensitivity; number of selections. "*" marks a cell that holds with a point estimate above nominal; "(abst.)" marks a pass by abstention; † marks a site seen in FLAMeS training.

design · test site	P1, α = 0.1 (FDR)	P1, α = 0.2 (FDR)	P2, γ = 0.1 (FDX vs δ = 0.1)	P2, γ = 0.2 (FDX vs δ = 0.1)
1-mm grid → MS3SEG	0.107 [0.075, 0.145] holds*; sens. 0.258; n = 957	0.123 [0.099, 0.151] holds; sens. 0.404; n = 1545	0.21 [0.13, 0.30] **violated**; sens. 0.121; n = 436	0.12 [0.06, 0.20] inconclusive; sens. 0.121; n = 436
1-mm grid → Muslim	0.122 [0.062, 0.197] inconclusive; sens. 0.095; n = 195	0.097 [0.058, 0.146] holds; sens. 0.235; n = 471	0.29 [0.17, 0.44] **violated**; sens. 0.036; n = 81	0.27 [0.16, 0.42] **violated**; sens. 0.036; n = 81
1-mm grid → PediMS	0.041 [0.009, 0.102] holds; sens. 0.120; n = 139	0.086 [0.056, 0.122] holds; sens. 0.326; n = 399	0.07 [0.00, 0.24] inconclusive; sens. 0.053; n = 60	0.07 [0.00, 0.24] inconclusive; sens. 0.053; n = 60
leave-one-site-out · MSLesSeg	0.009 [0.000, 0.029] holds (abst.); sens. 0.010; n = 34	0.018 [0.004, 0.042] holds; sens. 0.089; n = 312	0.02 [0.00, 0.06] holds (abst.); sens. 0.001; n = 4	0.03 [0.00, 0.07] holds (abst.); sens. 0.004; n = 16
leave-one-site-out · Ljubljana-10 †	0.000 [0.000, 0.000] holds (abst.); sens. 0.009; n = 6	0.011 [0.000, 0.033] holds; sens. 0.118; n = 80	0.00 [0.00, 0.31] inconclusive (abst.); sens. 0.011; n = 7	0.00 [0.00, 0.31] inconclusive (abst.); sens. 0.011; n = 7
leave-one-site-out · MS3SEG	0.087 [0.059, 0.121] holds; sens. 0.201; n = 744	0.116 [0.093, 0.142] holds; sens. 0.381; n = 1453	0.16 [0.09, 0.25] inconclusive; sens. 0.087; n = 316	0.12 [0.06, 0.20] inconclusive; sens. 0.087; n = 316
leave-one-site-out · Muslim	0.082 [0.033, 0.137] holds (abst.); sens. 0.030; n = 66	0.100 [0.048, 0.163] holds; sens. 0.139; n = 276	0.18 [0.08, 0.31] inconclusive (abst.); sens. 0.015; n = 37	0.18 [0.08, 0.31] inconclusive (abst.); sens. 0.015; n = 37
leave-one-site-out · PediMS	0.007 [0.000, 0.022] holds; sens. 0.064; n = 73	0.050 [0.016, 0.093] holds; sens. 0.164; n = 195	0.00 [0.00, 0.12] holds (abst.); sens. 0.025; n = 28	0.00 [0.00, 0.12] holds (abst.); sens. 0.025; n = 28



† Ljubljana is a FLAMeS training site: its 10 patients were never trained on, but the site was. 95 % patient-cluster bootstrap intervals (P2: union with exact Clopper–Pearson).

Figure 1. The 32 pre-registered primary cells. Left: realised FDR of conformal BH (P1). Right: realised FDX of the CALM-FDX threshold (P2). Shapes and colours encode the pre-registered class. Hollow markers pass only by abstention. The grey band is the pre-registered margin (nominal to nominal + 0.05).

No P1 cell was violated, but this is weaker than "the guarantee held". Across the 16 held-out-site P1 cells, none had a lower bound above α. Fourteen of these cells are on sites FLAMeS never trained on. The pre-registered verdict for thin-to-thick transport of P1 is nevertheless *inconclusive*, for four reasons:

1. On Muslim at α = 0.1 the result is undecided: 0.122 [0.062, 0.197].
2. Three of the 16 cells pass only by abstention: leave-one-site-out MSLesSeg, Ljubljana-10 and Muslim at α = 0.1.
3. Ljubljana-10 (10 patients) and PediMS (9) are small. Percentile bootstraps with so few clusters under-cover.
4. On MS3SEG at α = 0.1, the estimate (0.107 [0.075, 0.145]) is above α and the cell passes only through the pre-registered margin.

With Bonferroni intervals, no P1 cell is violated. The two-level bootstrap gave 0.101 [0.070, 0.136] (MS3SEG) and 0.104 [0.042, 0.182] (Muslim) at α = 0.1.

The calibration does not transfer as such. With within-site calibration, realised FDR at α = 0.1 was 0.041 on MS3SEG and 0.038 on Muslim. These values are close to α·π0 = 0.034 and 0.037 **[post hoc]**, where π0 is the share of false candidates. Conformal BH attains exactly this value under exchangeability. With 1-mm calibration, realised FDR was 0.107 and 0.122, that is 2.6–3.2 times higher. The thick-slice false candidates are therefore not exchangeable with the 1-mm null. The average bound survives only because many thick-slice scans are lesion-rich, so their false selections are diluted by true ones.

[post hoc] On the third of MS3SEG + Muslim scans with at least 40 % false candidates (51 of 151 scans; mean π0 0.52), realised FDR at α = 0.1 was **0.196 [0.120, 0.282]** with 1-mm calibration, against 0.050 [0.011, 0.103] with within-site calibration. With at least 50 % false candidates (24 scans) it was 0.225 [0.101, 0.366]. A marginal guarantee averages over a mixture of scans; for low-lesion-load thick-slice scans, the 1-mm calibration does not deliver it.

In the opposite direction, foreign calibration made the 1-mm sites nearly silent. On MSLesSeg at α = 0.1, leave-one-site-out calibration selected 34 candidates (sensitivity 0.010); within-site calibration selected 358 (sensitivity 0.102).

Pre-specified sensitivity analyses. Strict matching (IoU ≥ 0.10) and removing the PediMS patient with Baló's concentric sclerosis left every primary class unchanged. With the fold-consensus score S_cons, every D3 P1 cell held (Muslim α = 0.1: 0.068 [0.026, 0.122]) and no P2 cell was violated (MS3SEG γ = 0.1: 0.14 [0.08, 0.22], inconclusive). The power cost was modest: at α = 0.2, sensitivity on the thick sites was 0.296 with S_cons and 0.341 with the ensemble mean.

6.3 Per-scan control (P2) does not transport

With the threshold calibrated on the 1-mm sites ($\hat{\lambda}$ = 0.965), **21 % [13, 30] of MS3SEG scans (γ = 0.1) and 29 % [17, 44] / 27 % [16, 42] of Muslim scans (γ = 0.1 / 0.2) had FDP above γ**, against a budget of 10 %. The pre-registered verdict for Q1/P2 is *refuted*. Two qualifications apply:

- After Bonferroni correction for 32 cells, only Muslim at γ = 0.1 stays violated ([0.12, 0.52]).
- In the two-level bootstrap, the MS3SEG lower bound falls to the nominal 0.10 (0.20 [0.10, 0.30]), so that violation is borderline. The Muslim violations remain (0.27 [0.12, 0.41] and 0.25 [0.12, 0.37]).

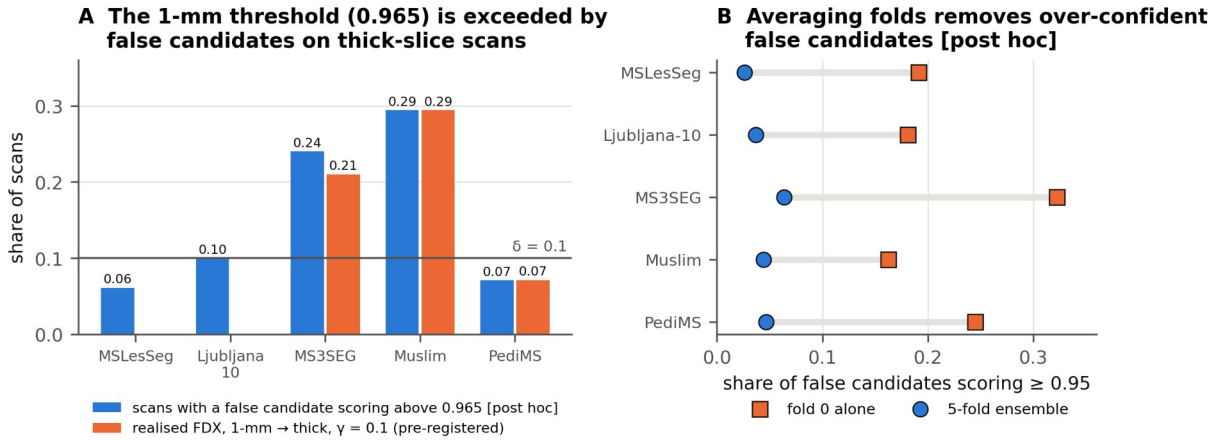


Figure 2. Why the per-scan threshold fails on thick slices. (A) Share of scans with a false candidate above the 1-mm-calibrated threshold 0.965 [post hoc], next to the pre-registered realised FDX of 1-mm → thick transport. (B) Share of false candidates scoring at least 0.95, fold 0 versus 5-fold ensemble [post hoc].

Mechanism [post hoc]. P2 reduces to a single global threshold. The share of scans with a false candidate above 0.965 was 6.1 % on MSLesSeg and 10 % on Ljubljana-10, but 24 % on MS3SEG and 29.4 % on Muslim. These shares nearly equal the realised FDX (Figure 2A). The false-score law of thick-slice scans differs from the 1-mm law. The pre-registered scan-clustered permutation KS test gave D = 0.107 on MS3SEG (p = 0.0005, the smallest attainable with 2000 permutations), D = 0.094 on Muslim (p = 0.025) and D = 0.057 on PediMS (p = 0.51).

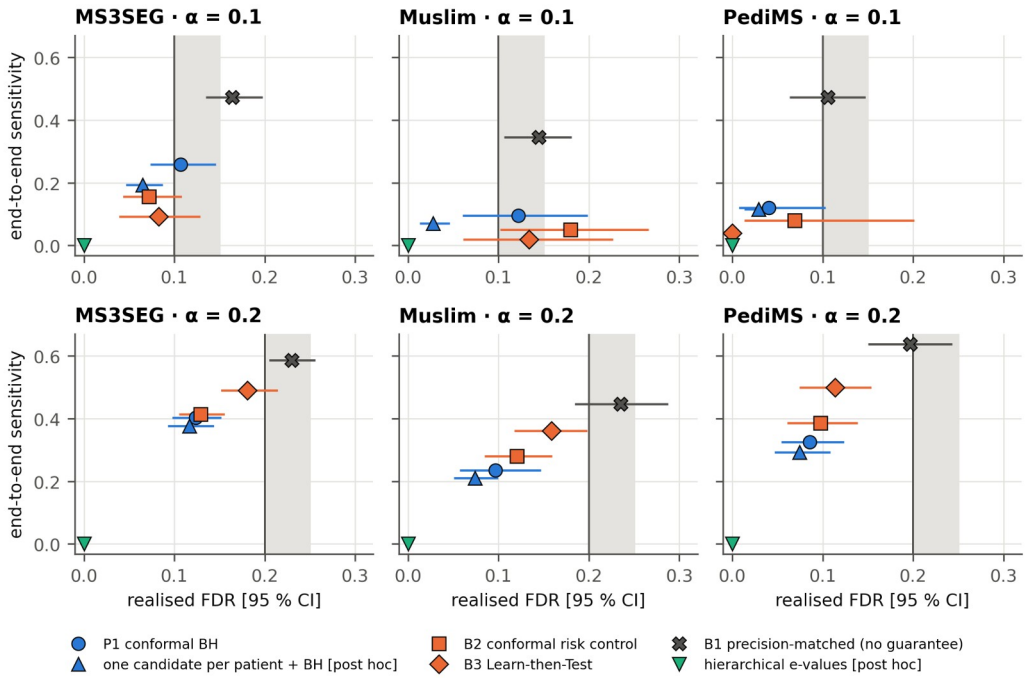
A single threshold carries this heavier upper tail straight into exceedance. P1 calibrates against the whole false-score law and controls only an expectation, which absorbs the shift with lost slack. At γ ≤ 0.2, P2 hardly depends on γ: for 22–51 % of calibration scans, the safe threshold lies above the scan's top false candidate. P2 therefore behaves like the rule "no false lesion above the threshold in 90 % of patients".

6.4 Baselines

Table 5. All procedures under 1-mm → thick transport (D3), ensemble arm. Entries: realised FDR [95 % CI] → class; end-to-end sensitivity. SUB entries average over 200 subsampling draws.

procedure	α	MS3SEG	Muslim	PediMS
P1 conformal BH (pre-registered, primary)	0.1	0.107 [0.075, 0.145] holds; sens. 0.258	0.122 [0.062, 0.197] inconclusive; sens. 0.095	0.041 [0.009, 0.102] holds; sens. 0.120
P1 conformal BH (pre-registered, primary)	0.2	0.123 [0.099, 0.151] holds; sens. 0.404	0.097 [0.058, 0.146] holds; sens. 0.235	0.086 [0.056, 0.122] holds; sens. 0.326
one false candidate per calibration patient + BH [post hoc]	0.1	0.065 [0.048, 0.086] holds; sens. 0.193	0.028 [0.014, 0.045] holds; sens. 0.071	0.029 [0.014, 0.047] holds; sens. 0.116

one false candidate per calibration patient + BH [post hoc]	0.2	0.117 [0.094, 0.143] holds; sens. 0.376	0.074 [0.052, 0.098] holds; sens. 0.210	0.075 [0.048, 0.107] holds; sens. 0.292
hierarchical conformal e-values + e-BH [post hoc]	0.1	no selection (sens. 0)	no selection (sens. 0)	no selection (sens. 0)
hierarchical conformal e-values + e-BH [post hoc]	0.2	no selection (sens. 0)	no selection (sens. 0)	no selection (sens. 0)
B2 conformal risk control, per-scan FDP loss (pre-registered)	0.1	0.072 [0.044, 0.107] holds; sens. 0.155	0.180 [0.103, 0.265] **violated**; sens. 0.050	0.069 [0.015, 0.200] inconclusive; sens. 0.080
B2 conformal risk control, per-scan FDP loss (pre-registered)	0.2	0.129 [0.106, 0.155] holds; sens. 0.414	0.120 [0.086, 0.158] holds; sens. 0.280	0.098 [0.062, 0.137] holds; sens. 0.386
B3 Learn-then-Test, per-scan FDP loss, δ = 0.1 (pre-registered)	0.1	0.083 [0.040, 0.128] holds; sens. 0.092	0.134 [0.062, 0.226] inconclusive; sens. 0.019	0.000 [0.000, 0.000] holds; sens. 0.039
B3 Learn-then-Test, per-scan FDP loss, δ = 0.1 (pre-registered)	0.2	0.181 [0.153, 0.213] holds; sens. 0.490	0.159 [0.119, 0.197] holds; sens. 0.361	0.114 [0.076, 0.152] holds; sens. 0.499
B1 precision-matched threshold, no guarantee (pre-registered)	0.1	0.164 [0.136, 0.196] **violated**; sens. 0.474	0.144 [0.107, 0.180] **violated**; sens. 0.346	0.106 [0.065, 0.146] holds; sens. 0.473
B1 precision-matched threshold, no guarantee (pre-registered)	0.2	0.230 [0.206, 0.255] **violated**; sens. 0.586	0.235 [0.186, 0.286] inconclusive; sens. 0.446	0.197 [0.152, 0.242] holds; sens. 0.639



Calibration on the 1-mm-grid sites, test on each thick-slice site (D3). Hierarchical e-values selected nothing in every D3 cell (sensitivity 0). Grey band: nominal α to α + 0.05.

Figure 7. Validity against power for every procedure under 1-mm → thick transport. The vertical line is α; the grey band reaches α + 0.05.

Pre-registered baselines.

- **B1 (precision-matched threshold)** was violated on MS3SEG at both levels (α = 0.1: 0.164 [0.136, 0.196]; α = 0.2: 0.230 [0.206, 0.255]) and on Muslim at α = 0.1 (0.144 [0.107, 0.180]).
- **B2 (conformal risk control)** was violated on Muslim at α = 0.1 (0.180 [0.103, 0.265]).

- **B3 (Learn-then-Test, per-scan FDP loss)** had no violated D3 or D2 cell. Its pattern resembled P1, with Muslim at α = 0.1 inconclusive (0.134 [0.062, 0.226]). The two procedures distribute power differently. At α = 0.2, B3 found more lesions than P1 (0.490, 0.361 and 0.499 against 0.404, 0.235 and 0.326 on MS3SEG, Muslim and PediMS), with realised FDR closer to α (0.181, 0.159, 0.114). At α = 0.1, it found fewer (0.092, 0.019, 0.039).

Our earlier text stated that conformal BH was the only procedure without a violated thin-to-thick cell; that statement was wrong (Table 11). B3's higher power at α = 0.2 also came with a violated healthy-control cell (§6.9).

Post-hoc scan-aware baselines.

- **SUB** (one false candidate per calibration patient) had no violated cell in D3 or D2. On thick slices its averaged realised FDR was lower than P1's (MS3SEG α = 0.1: 0.065 [0.048, 0.086]; Muslim: 0.028 [0.014, 0.045]), at a cost in power at α = 0.1 (MS3SEG 0.193 against 0.258). However, a deployment uses a single subsampling draw, and single draws varied widely. On MS3SEG at α = 0.1, the central 95 % of draws ranged from 0.004 to 0.115, and 14.5 % of draws exceeded α.
- **LR-hier** (hierarchical e-values) selected nothing in every D3 cell. With K = 85 calibration patients its e-values are at most K + 1 = 86, while e-BH needs k e-values of at least n/(α·k) to select k of n test candidates. With n in the thousands, it must select hundreds at once or none. In D2, with 145–236 calibration patients, it selected in 3 of 10 cells and then selected heavily. On MS3SEG at α = 0.2, the batch FDP was 0.202 and the mean per-scan FDP 0.212 [0.188, 0.239], at sensitivity 0.559. Its behaviour is all-or-nothing.

D2 results for all procedures are in Appendix Table A2.

6.5 Ljubljana removed from every calibration pool [post hoc]

Ljubljana-10 contributed 82 of the 1339 false candidates in the 1-mm null. Removing Ljubljana from every pool changed no class in D3 or D2 (Appendix Table A3). On MS3SEG at α = 0.1, the D3 estimate moved from 0.107 to 0.098 [0.071, 0.131]. Muslim at α = 0.1 stayed at 0.122 [0.062, 0.197]. Every D3 P2 cell was identical, because the MSLesSeg-only threshold is also 0.965. The seen-site calibration data do not drive the results.

6.6 Slice thickness — a gradient confounded with site

Table 6. Validity and power by thickness band, leave-one-site-out calibration (D2), ensemble arm. Bands were fixed before outcomes. Each band is essentially one or two sites.

band (slice spacing) · sites	scans / patients	P1 α = 0.1: FDR · sens.	P1 α = 0.2: FDR [95 % CI]	scans with FDP > 0.1 (α = 0.2)	P1 α = 0.2: sens. [95 % CI]	false / true selected lesions per scan (α = 0.2)	P2 γ = 0.1: FDX [95 % CI] · sens.
≤ 1.5 mm grid (native: 10 at 0.8 mm, 110 at 2–5 mm, 5 unknown) — MSLesSeg, Ljubljana-10	125 / 85	0.008 · 0.010	0.018 [0.004, 0.040]	6 %	0.094 [0.053, 0.141]	0.13 / 3.0	0.016 [0.000, 0.057] · 0.002
(1.5, 4.5] mm: 3.9 mm — PediMS	28 / 9	0.007 · 0.064	0.050 [0.016, 0.093]	14 %	0.164 [0.042, 0.440]	0.50 / 6.5	0.000 [0.000, 0.123] · 0.025
(4.5, 6.5] mm — Muslim 45, MS3SEG 31	76 / 76	0.084 · 0.112	0.101 [0.063, 0.145]	30 %	0.236 [0.180, 0.298]	0.88 / 7.9	0.158 [0.079, 0.260] · 0.047
> 6.5 mm — MS3SEG 69, Muslim 6	75 / 75	0.087 · 0.170	0.120 [0.093, 0.152]	49 %	0.355 [0.310, 0.402]	1.73 / 12.4	0.173 [0.093, 0.278] · 0.077

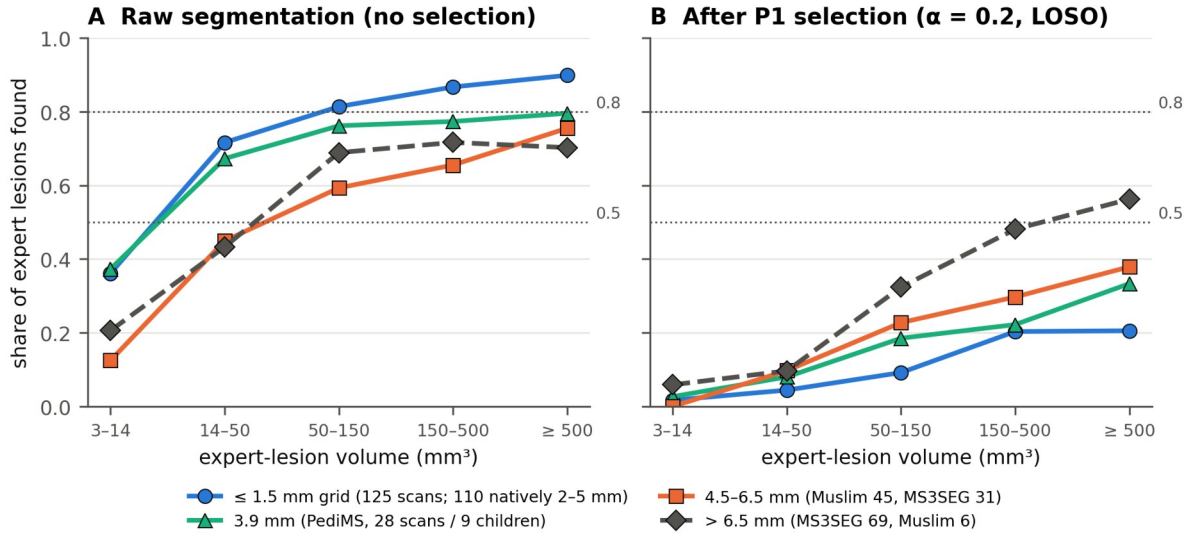
Across bands, no P1 cell showed a violation. As slices thicken, the slack shrinks and per-scan exceedance grows. At α = 0.2, realised FDR rose from 0.018 (≤ 1.5 mm grid) to 0.050 (3.9 mm), 0.101 (4.5–6.5 mm) and 0.120 (> 6.5 mm). The share of scans with FDP > 0.1 rose from 6 % to 14 %, 30 % and 49 %. The pooled design (D4) showed the same ordering.

This is mostly a between-site difference, not a demonstrated thickness effect. Within sites, the gradient was small. On MS3SEG, realised FDR went from 0.103 (4.5–6.5 mm) to 0.121 (> 6.5 mm). On Muslim it went from 0.100 to 0.102, with only 6 scans above 6.5 mm. **[post hoc]** Within MS3SEG (6.0–8.1 mm), per-scan FDP did not trend with slice spacing (Spearman ρ = 0.03, p = 0.75); on Muslim, ρ = 0.30 (p = 0.035).

Power at the guarantee is higher on thick-slice sites. **[post hoc]** Their expert lesions are larger: median 108–117 mm³, against 68 mm³ on MSLesSeg and 20 mm³ on Ljubljana. The 1-mm band is also penalised by the foreign null.

Table 7. Expert-lesion sensitivity by lesion volume and band (raw = matched by any candidate, before selection; at the guarantee = selected by P1 at α = 0.2 in D2).

band	lesions per bin (3–14 / 14–50 / 50–150 / 150–500 / ≥ 500 mm³)	raw sensitivity per bin	sensitivity at P1 α = 0.2 (leave-one-site-out)	raw ≥ 0.8 from	raw ≥ 0.5 from
≤ 1.5 mm grid	531 / 1217 / 1327 / 610 / 326	0.36 / 0.72 / 0.81 / 0.87 / 0.90	0.02 / 0.04 / 0.09 / 0.20 / 0.21	50 mm³	14.14 mm³
3.9 mm	75 / 324 / 378 / 234 / 93	0.37 / 0.67 / 0.76 / 0.77 / 0.80	0.03 / 0.08 / 0.19 / 0.22 / 0.33	not reached	14.14 mm³
4.5–6.5 mm	48 / 514 / 976 / 673 / 343	0.13 / 0.45 / 0.59 / 0.66 / 0.76	0.00 / 0.10 / 0.23 / 0.30 / 0.38	not reached	50 mm³
> 6.5 mm	34 / 455 / 1074 / 713 / 343	0.21 / 0.43 / 0.69 / 0.72 / 0.70	0.06 / 0.10 / 0.32 / 0.48 / 0.56	not reached	50 mm³
within MSLesSeg, native 2–2.3 mm [post hoc]	141 / 587 / 594 / 274 / 135	0.37 / 0.74 / 0.86 / 0.91 / 0.98	—	—	—
within MSLesSeg, native 3 mm [post hoc]	94 / 305 / 435 / 210 / 132	0.29 / 0.65 / 0.77 / 0.83 / 0.86	—	—	—
within MSLesSeg, native 4–5 mm [post hoc]	25 / 59 / 143 / 71 / 35	0.16 / 0.39 / 0.73 / 0.82 / 0.71	—	—	—



Bands are confounded with site, scanner, population and annotation protocol; they describe these cohorts, not a controlled thickness effect.

Figure 3. Share of expert lesions found by lesion volume and thickness band: (A) raw segmentation, (B) after P1 selection at α = 0.2, leave-one-site-out.

On 2D slices of 4.5 mm or more, before any selection:

- lesions of at least 50 mm³ (about 4.6 mm across) were found 59–76 % of the time, depending on the size bin;
- lesions of 14–50 mm³ were found 43–45 % of the time;
- lesions under 14 mm³ were found 13–21 % of the time;
- no lesion size reached 80 %.

On the 1-mm grid, 80 % was reached from 50 mm³. At the guarantee, sensitivity exceeded one half only for lesions of at least 500 mm³ on slices thicker than 6.5 mm. The PediMS band missed 0.8 by a small margin on 93 lesions (≥ 500 mm³: 0.796).

[post hoc] Thickness matters even inside the "1-mm" band. Within MSLesSeg, raw detection of 14–50 mm³ lesions fell with native thickness: 0.74 at 2–2.3 mm (43 scans), 0.65 at 3 mm (52) and 0.39 at 4–5 mm (15). This within-site gradient is confounded with scanner and patient.

Muslim's estimated pixel size moves its own 50 %-detection size between 50 and 150 mm³: at a volume scale of 1.1, its 50–150 mm³ bin falls to 0.487.

6.7 The 5-fold ensemble against a single fold

Table 8. Ensemble against fold 0 on the same scans and expert lesions (P1; patient-paired bootstrap).

design · test set	α	realised FDR: ensemble / fold 0	Δ FDR [95 % CI]	sensitivity: ensemble / fold 0	Δ sensitivity [95 % CI]
1-mm → thick, all thick sites	0.2	0.110 / 0.178	-0.068 [-0.098, -0.037]	0.341 / 0.194	+0.147 [0.117, 0.177]
1-mm → MS3SEG	0.2	0.123 / 0.225	-0.102 [-0.131, -0.074]	0.404 / 0.282	+0.121 [0.087, 0.153]
1-mm → Muslim	0.2	0.097 / 0.122	-0.025 [-0.104, 0.053]	0.235 / 0.080	+0.155 [0.096, 0.216]
1-mm → PediMS	0.2	0.086 / 0.110	-0.024 [-0.094, 0.030]	0.326 / 0.112	+0.214 [0.143, 0.240]
leave-one-site-out, all sites	0.2	0.067 / 0.084	-0.018 [-0.034, -0.000]	0.203 / 0.102	+0.101 [0.074, 0.129]
within-site, all sites	0.2	0.061 / 0.085	-0.025 [-0.045, -0.004]	0.219 / 0.078	+0.141 [0.108, 0.178]
pooled 5-fold, all sites	0.2	0.063 / 0.070	-0.008 [-0.025, 0.011]	0.196 / 0.069	+0.126 [0.096, 0.157]
1-mm → thick, all thick sites	0.1	0.101 / 0.136	-0.035 [-0.066, -0.003]	0.187 / 0.111	+0.076 [0.050, 0.106]
1-mm → MS3SEG	0.1	0.107 / 0.201	-0.095 [-0.132, -0.058]	0.258 / 0.174	+0.084 [0.048, 0.119]
1-mm → Muslim	0.1	0.122 / 0.052	+0.070 [0.012, 0.143]	0.095 / 0.035	+0.061 [0.028, 0.095]
1-mm → PediMS	0.1	0.041 / 0.057	-0.016 [-0.061, 0.058]	0.120 / 0.044	+0.076 [0.018, 0.219]
leave-one-site-out, all sites	0.1	0.046 / 0.068	-0.021 [-0.040, -0.001]	0.082 / 0.054	+0.028 [0.016, 0.042]
within-site, all sites	0.1	0.028 / 0.037	-0.008 [-0.026, 0.010]	0.064 / 0.027	+0.036 [0.022, 0.053]
pooled 5-fold, all sites	0.1	0.038 / 0.040	-0.002 [-0.018, 0.016]	0.063 / 0.019	+0.044 [0.030, 0.059]

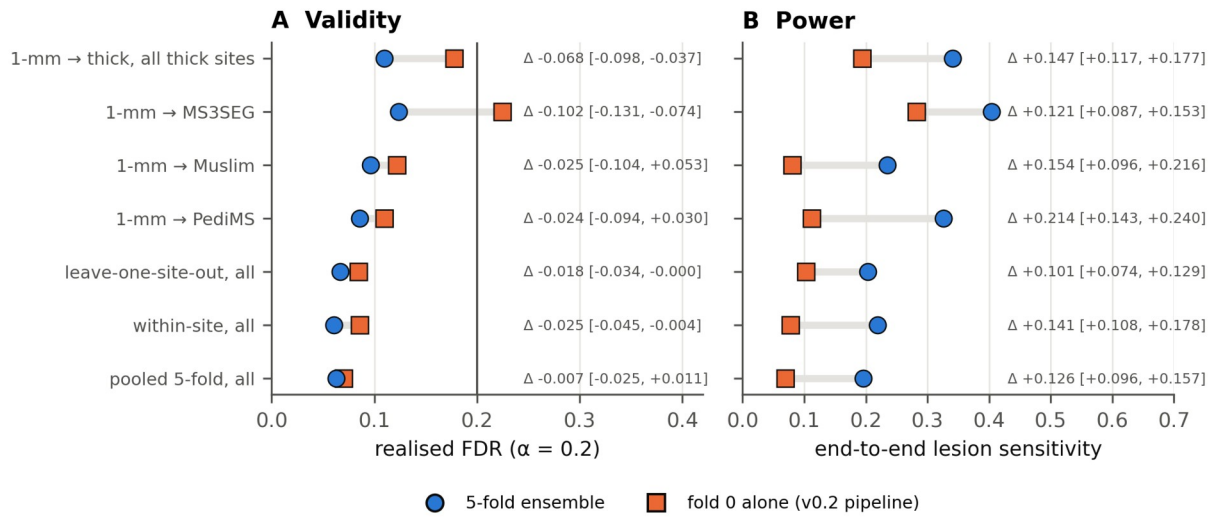


Figure 4. 5-fold ensemble against fold 0, P1 at α = 0.2: (A) realised FDR, (B) end-to-end sensitivity, with paired differences and 95 % CIs.

At α = 0.2, the ensemble roughly doubled end-to-end sensitivity at the guarantee:

- leave-one-site-out: 0.203 against 0.102 (Δ +0.101 [0.074, 0.129]);
- 1-mm → thick: 0.341 against 0.194 (Δ +0.147 [0.117, 0.177]).

Pooled realised FDR was lower (1-mm → thick: 0.110 against 0.178). At α = 0.1, the gain was much smaller: leave-one-site-out sensitivity went from 0.054 to 0.082 (Δ +0.028 [0.016, 0.042]).

Validity was not uniformly better. With fold 0 alone (our earlier pipeline), MS3SEG was violated in several cells:

- 1-mm → thick, α = 0.1: 0.201 [0.166, 0.242];
- 1-mm → thick, α = 0.2: 0.225 [0.193, 0.260];
- leave-one-site-out, α = 0.1: 0.176 [0.139, 0.216];
- every MS3SEG P2 cell (FDX 0.39–0.62).

The ensemble removed these five violated cells. On Muslim, however, the ensemble **raised** realised FDR. With 1-mm → thick calibration at α = 0.1 it reached 0.122 against 0.052 (Δ +0.070 [0.012, 0.143]). With leave-one-site-out calibration it reached 0.082 against 0.020 at α = 0.1 (Δ [0.017, 0.121]) and 0.100 against 0.044 at α = 0.2 (Δ [0.006, 0.116]). It also created two new violated cells (Muslim, P2). In these Muslim comparisons fold 0 had selected almost nothing (sensitivity 0.002–0.035), so its low FDR partly reflects abstention. The pre-registered verdict is "helps on power, but with new violated cells". Folds 1–4 behaved like fold 0 (1-mm → thick, P1 α = 0.2: realised FDR 0.157–0.176), so the gain comes from averaging, not from a bad fold.

[post hoc] Averaging removes over-confident false candidates. The share of false candidates scoring at least 0.95 fell from 16–32 % (fold 0) to 3–6 % (ensemble) at every site (Figure 2B).

LST-AI replicated the direction through a different mechanism. Its three-model ensemble did not rank better than its first sub-model (MSLesSeg AUROC 0.737 against 0.758). It did produce far fewer false candidates (candidate precision 0.45 against 0.25), so its within-site sensitivity at α = 0.2 was higher (0.102 against 0.017).

6.8 Local recalibration

Within-site calibration kept P1 below α at every site (Appendix Table A1). At α = 0.2, realised FDR was 0.034–0.075 and sensitivity 0.14–0.31.

Table 9. Recalibration with k local patients (D5; mean over 200 draws).

site · procedure · mode	k = 3	k = 10	k = 40	k*
MS3SEG · P1 α = 0.2 · site patients only	FDR 0.056 (0 % of draws > nominal); sens. 0.166	FDR 0.069 (0 % of draws > nominal); sens. 0.183	FDR 0.078 (0 % of draws > nominal); sens. 0.193	10
MS3SEG · P1 α = 0.1 · 1-mm pool + k site patients	FDR 0.104 (72 % of draws > nominal); sens. 0.251	FDR 0.101 (56 % of draws > nominal); sens. 0.233	FDR 0.081 (14 % of draws > nominal); sens. 0.185	20
MS3SEG · P2 γ = 0.1 · site patients only	abstains (k < 9)	FDX 0.075 (30 % of draws > nominal); sens. 0.039	FDX 0.086 (33 % of draws > nominal); sens. 0.038	10
MS3SEG · P2 γ = 0.1 · 1-mm pool + k site patients	FDX 0.192 (100 % of draws > nominal); sens. 0.110	FDX 0.187 (100 % of draws > nominal); sens. 0.105	FDX 0.162 (86 % of draws > nominal); sens. 0.088	none
Muslim · P1 α = 0.2 · site patients only	FDR 0.045 (0 % of draws > nominal); sens. 0.131	FDR 0.053 (0 % of draws > nominal); sens. 0.150	FDR 0.068 (3 % of draws > nominal); sens. 0.159	10
Muslim · P1 α = 0.1 · 1-mm pool + k site patients	FDR 0.103 (51 % of draws > nominal); sens. 0.092	FDR 0.095 (39 % of draws > nominal); sens. 0.086	FDR 0.076 (36 % of draws > nominal); sens. 0.066	5
Muslim · P2 γ = 0.1 · site patients only	abstains (k < 9)	FDX 0.092 (36 % of draws > nominal); sens. 0.007	FDX 0.096 (26 % of draws > nominal); sens. 0.005	10
Muslim · P2 γ = 0.1 · 1-mm pool + k site patients	FDX 0.250 (100 % of draws > nominal); sens. 0.027	FDX 0.235 (100 % of draws > nominal); sens. 0.023	FDX 0.196 (70 % of draws > nominal); sens. 0.017	none

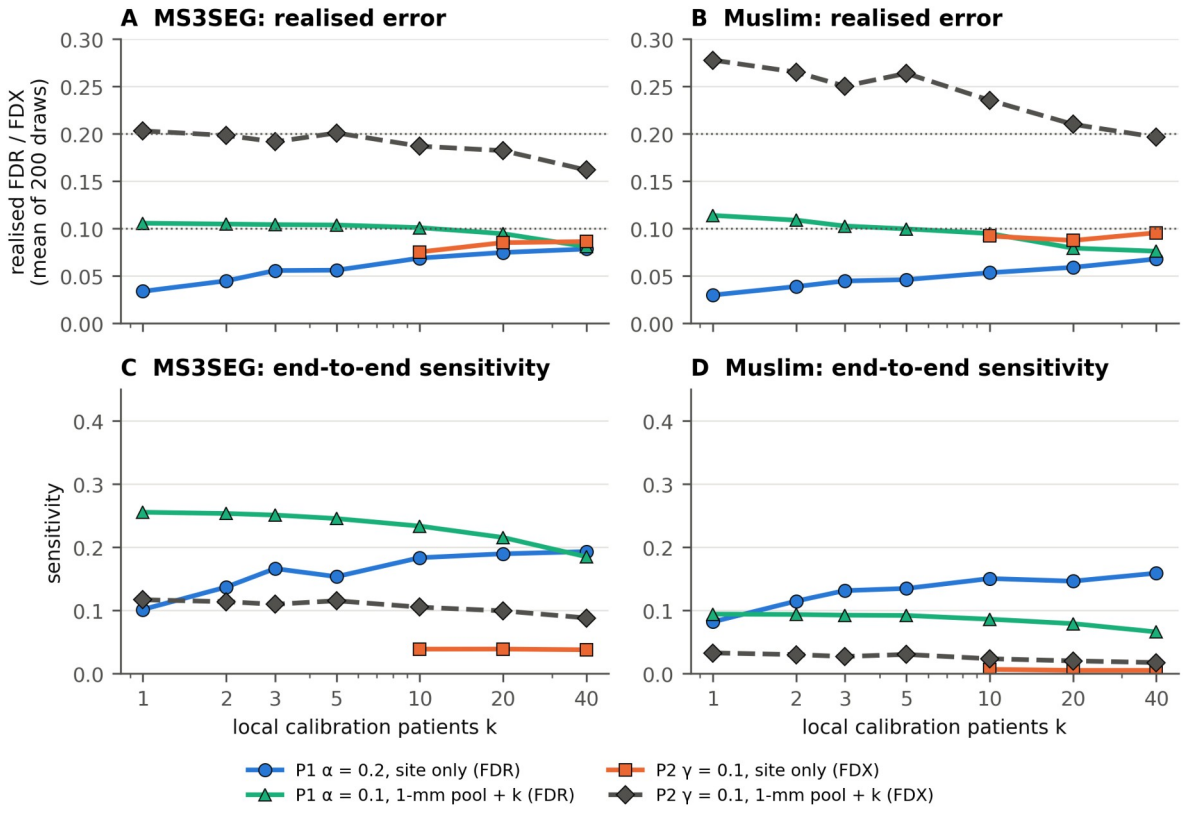


Figure 5. Recalibration with k local patients on MS3SEG and Muslim: realised error (top) and end-to-end sensitivity (bottom).

P1, site patients only. P1 stayed below α at every k. With 10 local patients it reached about 90 % of the full within-site power (MS3SEG α = 0.2: realised FDR 0.069, sensitivity 0.183 against 0.198 within site; Muslim: 0.053, 0.150 against 0.163).

P2 after recalibration — valid, but nearly silent. P2 cannot select with fewer than 9 calibration patients (δ = 0.1). With 10 local patients, its mean exceedance was back within budget (MS3SEG 0.075, Muslim 0.092), but **it found 3.85 % (MS3SEG) and 0.65 % (Muslim) of lesions**. Calibration-conditional failures remained common: in 30 % and 36 % of draws, FDX exceeded 0.1, as expected for a marginal guarantee with 10 calibration units (Vovk, 2013). The pre-registered criterion formally declares the violated P2 cells "restored with k* = 10"; in practice this is restoration by near-abstention.

Augmented pools. Adding local patients to the 85-patient 1-mm pool never restored P2 (FDX 0.16–0.28 for k ≤ 40) and left P1 at α = 0.1 on the edge, because the foreign pool dominates the quantile.

6.9 Healthy controls (separately pre-registered)

On a lesion-free scan, every candidate is false, and the promise of P1 reads "P(at least one reported lesion) ≤ α", provided that healthy candidates are exchangeable with the patients' false candidates. The promise of P2 reads "≤ δ = 0.1". Calibrations were frozen from patient data before any control outcome.

Table 10. Healthy controls (of 100) with at least one reported lesion [exact 95 % CI, %] → class. Primary cells are in bold.

arm · calibration pool	P1 α = 0.1	P1 α = 0.2	P2 γ = 0.1 (= 0.2)	B2 CRC α = 0.1 / 0.2	B3 LTT α = 0.1 / 0.2
ensemble · 1-mm-grid patient sites (primary)	**14 [7.9, 22.4] inconclusive**	**24 [16.0, 33.6] inconclusive**	**6 [2.2, 12.6] holds**	7 holds / 25 inconclusive	2 holds / 32 **violated**
ensemble · MSLesSeg only	15 [8.6, 23.5] inconclusive	24 [16.0, 33.6] inconclusive	6 [2.2, 12.6] holds	7 holds / 25 inconclusive	2 holds / 30 **violated**
ensemble · all five patient sites	5 [1.6, 11.3] holds	19 [11.8, 28.1] inconclusive	2 [0.2, 7.0] holds	2 holds / 11 holds	2 holds / 32 **violated**
ensemble · thick-slice sites	3 [0.6, 8.5] holds	14 [7.9, 22.4] holds	2 [0.2, 7.0] holds	2 holds / 7 holds	1 holds / 29 **violated**
ensemble · own null (leave-one-control-out; exchangeable by construction)	4 [1.1, 9.9]	13 [7.1, 21.2]	—	—	—
fold 0 · 1-mm-grid patient sites	59 [48.7, 68.7] **violated**	72 [62.1, 80.5] **violated**	32 [23.0, 42.1] **violated**	38 **violated** / 53 **violated**	14 inconclusive / 50 **violated**
fold 0 · MSLesSeg only	59 [48.7, 68.7] **violated**	72 [62.1, 80.5] **violated**	33 [23.9, 43.1] **violated**	38 **violated** / 53 **violated**	14 inconclusive / 48 **violated**
fold 0 · all five patient sites	39 [29.4, 49.3] **violated**	58 [47.7, 67.8] **violated**	7 [2.9, 13.9] holds	13 inconclusive / 32 **violated**	8 inconclusive / 48 **violated**
fold 0 · thick-slice sites	27 [18.6, 36.8] **violated**	51 [40.8, 61.1] **violated**	4 [1.1, 9.9] holds	8 inconclusive / 20 inconclusive	4 holds / 19 inconclusive
fold 0 · own null (leave-one-control-out; exchangeable by construction)	9 [4.2, 16.4]	21 [13.5, 30.3]	—	—	—

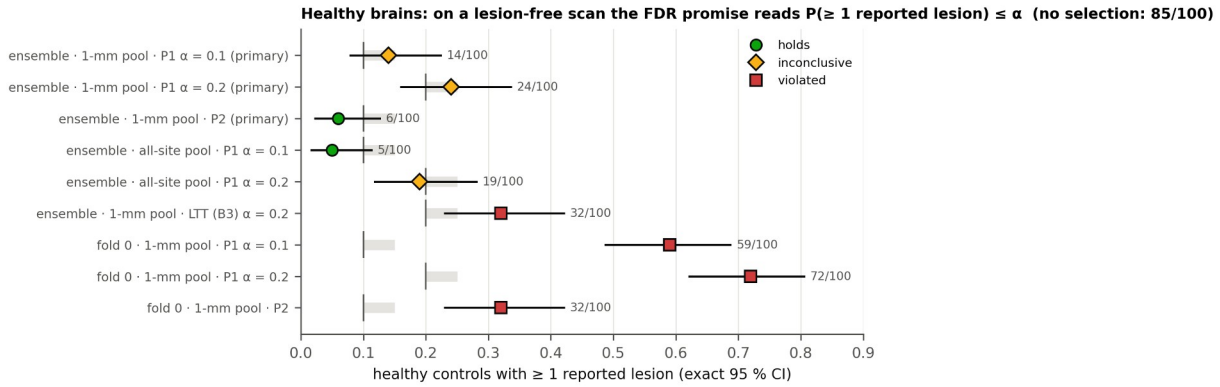


Figure 6. Healthy controls with at least one reported lesion, by arm, calibration pool and procedure (exact 95 % CIs). Grey band: nominal to nominal + 0.05.

Primary result. With the calibration a deployer would use for 1-mm-grid studies, the healthy false-alarm rate was undecided and above nominal. **14/100 [7.9, 22.4 %] controls had a reported lesion at α = 0.1 and 24/100 [16.0, 33.6 %] at α = 0.2**; both cells are *inconclusive*. P2 flagged 6/100 [2.2, 12.6 %] (holds), but it is the procedure that finds almost nothing in patients. Of the 70 controls never analysed before, 13/70 (18.6 % [10.3, 29.7]) were flagged at α = 0.1, a violated cell in that subset. Of the 30 analysed earlier, 1/30 was flagged (**[post hoc]** Fisher exact test, p = 0.06). Without the one control with a poor brain mask, the counts were 14/99 and 23/99.

The pool matters, in the expected direction. Pools that include thick-slice patients produced fewer healthy false alarms (all five sites: 5/100 at α = 0.1), because their false-score tail is heavier. **[post hoc]** The controls' candidates resemble the thick-slice null more than the 1-mm null: 5.7 % scored at least 0.95, against 2.7 % of the 1-mm null and 5.6 % of the thick-slice null. LTT (B3) at α = 0.2 was violated with every pool (29–32/100).

A single fold fails on healthy brains. Fold 0 with 1-mm calibration flagged 59/100 controls at α = 0.1 (violated). The ensemble flagged 14/100; the paired difference was −45 points [−55, −35], with 1 against 46 discordant controls (exact McNemar $p < 10^{-5}$).

Without selection, the segmentation marks most healthy brains. The ensemble produced at least one candidate in 85/100 controls (2.63 per scan); fold 0 did so in 93/100.

Dissemination in space on healthy brains. We ran the software's own brain-DIS code on these template-space images: MSMask atlas zones, the MAGNIMS contact rule, and at least 2 of the periventricular, juxtacortical and infratentorial regions. Spinal cord and optic nerve were not available. Results:

- Applied to the **unselected** segmentation, the rule was met in **19/100 healthy controls (11.8–28.1 %)**; fold 0 met it in 22/100.
- 79/100 controls had at least one automatic lesion in a DIS region.
- After selection, the rule was met in 2/100 (P1 α = 0.1), 3/100 (α = 0.2) and 0/100 (P2; one-sided 95 % bound 3.0 %).

The false lesions were not skull-stripping artefacts: 2 of 263 lay within 3 mm of the brain-mask edge. Of the 263, 165 (63 %) were periventricular; their median volume was 27 mm^3. **[post hoc]** On visual review, the top-scoring

false candidates looked like normal periventricular caps and small focal white-matter hyperintensities. No expert read these scans, so some may be genuine incidental findings.

Our earlier drafts claimed that no healthy control met DIS: first "0/200 (95 % CI ≤ 1.8 %)", which counted 20 resplits of the same 30 scans as independent, then "0/30 (≤ 9.5 %)". Both statements are withdrawn and replaced by these numbers.

6.10 Paired segmenter swap (exploratory)

On the same MSLesSeg images, we used 5-fold patient cross-fitting with calibration and test patients disjoint:

- FLAMeS-calibrated selection on LST-AI candidates selected nothing.
- The reverse, LST-AI calibration applied to FLAMeS candidates, gave realised FDR **0.207 [0.175, 0.240]** at α = 0.1 and 0.236 [0.203, 0.268] at α = 0.2. At α = 0.1, 78 % of scans had FDP > 0.1.
- Within-segmenter calibration was valid for both: FLAMeS 0.019 / 0.052 and LST-AI 0.053 / 0.095.

This failure was expected. The two segmenters' scores live on different scales (mean false-candidate score 0.65 for LST-AI against 0.76 for the FLAMeS ensemble), and the experiment covers one site. It illustrates a deployment rule: calibrate with the segmenter version that is deployed.

7 Translation into software: a default-off slice-thickness reliability card

The findings were turned into a software change in MSTool-AI, the author's MS imaging application, which is developed as IEC 62304 Class C software (International Electrotechnical Commission, 2015). The change went through that standard's change-control process (change proposal CP-SLICE-THICKNESS-001; requirements REQ-FUNC-098/099 and REQ-SAFE-034/035/036; risk controls RC-054 to RC-058; hazard HAZ-015). We describe it as **translation, not clinical validation**. The card has not been evaluated with readers, and its figures have not been re-derived for the deployed segmenter version.

The card sits next to the lesion results and does five things:

1. **It reads the slice thickness of the original acquisition and never assumes it.** It reports the thickness as native (from an original DICOM with recorded slice thickness), as a lower bound ("at least", from an image grid), or as not determinable. A 1-mm grid without acquisition metadata is "not determinable", because MSLesSeg shows that 2–5 mm acquisitions are routinely delivered on 1-mm grids (§4.1).

2. **It shows observed detection by lesion size** for the band, with its evidence size: Table 7, the pre-registered bands only, worst case within the band (and across thicker bands when only a lower bound on the thickness is known). The figures come from a versioned evidence asset bound by SHA-256 to the result record. Post-hoc rows are refused.

3. **It shows no false-lesion rate at any thickness**, and says why. On thick slices this follows §6.2–6.3: no figure without a calibration of at least 10 patients from the same site and protocol with the deployed segmenter (§6.8). No such site calibration exists in this version.

4. **It states that an empty or short list does not mean that there are no lesions**, and it recommends 3D FLAIR at 1.5 mm or less for small lesions on thick or unknown thickness.

5. **It states its scope**: research evidence from five public cohorts and one segmenter; not a diagnosis; not for counting lesions for the McDonald criteria; no statement about healthy or screening populations.

The card is behind a feature flag that is **off by default**. It may only be turned on after the figures have been re-derived for the deployed segmenter version and the wording has been reviewed with radiologists.

A related safety rule followed from §6.9: dissemination in space must not be computed from an automatic segmentation that a clinician has not reviewed. It was handled through the software's problem-reporting process and is outside the scope of this paper.

8 Discussion

8.1 What held and what failed

Three findings held.

- **Marginal lesion-wise FDR control (P1) showed no detected violation** on 16 held-out-site comparisons, including transport from 1-mm-grid to 4–8 mm 2D acquisitions. The same pattern held with the training site removed from calibration, with strict matching, with a consensus score and in the marginal (two-level) view.
- **LTT with a per-scan loss (B3)** showed the same pattern and had more power at $\alpha = 0.2$.
- **Within-site calibration** kept every P1 cell below α.

Five findings failed or remained unsettled.

1. **Per-scan control (P2) failed** when calibrated on thin and applied to thick slices. It was restored only by local calibration with at least 10 patients, at which point it found 0.65–3.85 % of lesions.

2. **Thin-slice calibration does not transfer as a calibration.** On thick slices, realised FDR was 2.6–3.2 times its within-site value. Post hoc, low-lesion-load scans received about twice the promised rate.

3. **One thick-slice site remained undecided** at $\alpha = 0.1$.

4. **The healthy-brain promise was undecided** with point estimates above nominal.

5. **Thickness was confounded with site** throughout. The cohorts cannot separate a thickness effect from scanner, population and annotation protocol.

8.2 Why the marginal promise survives and the per-scan promise does not

Under exchangeability, conformal BH attains $\alpha \cdot \pi_0$, about $\alpha/3$ here. That slack absorbs a moderate shift of the null. The realised FDR on thick slices (0.107 and 0.122 at $\alpha = 0.1$) roughly consumed it. Lesion-rich scans dilute their few false selections, so the average stays near α even when low-lesion-load scans exceed it.

The per-scan threshold has no slack to spend. Its realised exceedance equals the share of scans whose highest false candidate lies above one number, and thick-slice scans carry a heavier upper tail of confident false candidates. Two reasons follow.

- **The "guarantee held" in the marginal sense, but it is a property of the mixture of scans, not of the scan a reader is looking at.**
- **A marginal per-scan guarantee with few calibration units fails conditionally in a large share of calibration draws** (30–36 % with 10 patients).

This is the known gap between marginal and calibration-conditional validity (Vovk, 2013; Angelopoulos et al., 2025).

The scan-level random effect was modest: the intraclass correlation of false-candidate scores was 0.10 on MSLesSeg with the ensemble and below 0.03 elsewhere. The violations stem mainly from a shift in the law of false-candidate scores between acquisition types, not from within-scan clustering. Scan-aware procedures (B2, B3, SUB, LR-hier) do not repair a shift. They change which unit must be exchangeable, and they trade power in different ways.

8.3 Implications for deployment

- **Calibrate with the deployed segmenter and with data from the target site or acquisition type.** Calibration from another segmenter can double the false-discovery rate (§6.10). Calibration from thin slices does not carry over to thick ones (§6.2–6.3).
- **Ensemble the folds** for power. Do not expect the ensemble to fix validity: it fixed one site and worsened another (§6.7).
- **Do not offer per-scan promises** on this evidence. Where they are valid, they are nearly silent.
- **Do not extend patient-calibrated promises to healthy or screening populations.** Do not compute dissemination in space from unreviewed automatic lesions (§6.9).
- **Expect low power at the guarantee.** End-to-end sensitivity was 0.01–0.40 for P1 depending on site, design and α. The main lever is better candidate scores: the ensemble's AUROC gain doubled power without touching the wrapper.

8.4 Relation to prior work

Our transport failure for a per-scan quantity agrees with Adhikary et al. (2026), who found per-organ recall risk control in CT failing under clinical domain shift. Our analysis is lesion-wise, uses MS and thick-slice acquisitions, and adds the marginal-versus-per-scan contrast. Everywhere-valid and exact-law FDP bounds (Gazin et al., 2024; Song et al., 2026) assume candidate exchangeability, which our thick-slice data violate at the level of the score law. Weighted conformal selection (Jin and Candès, 2026) would need a likelihood ratio between acquisition types that is not available from anisotropic shape features. Mondrian calibration by acquisition type (Vovk et al., 2005) is the natural remedy that our recalibration curve supports, and it should be tested prospectively.

8.5 Corrections to our earlier drafts

Appendix B lists every claim of our earlier internal drafts that this audit retracts or changes. The two most consequential concern contamination and healthy controls.

- **Contamination.** The first draft evaluated FLAMeS on 20 of its own training scans.
- **Healthy controls.** The first draft reported "0 of 200 healthy scans certified as DIS (95 % CI ≤ 1.8 %)". The 200 were 20 resplits of 30 scans; the honest ceiling from 30 scans is about 10 %, and on 100 distinct controls the unselected segmentation met the DIS rule in 19 %.

9 Limitations

Segmenter. One segmenter family (FLAMeS) carries all multi-site results. LST-AI appears only on the 1-mm-grid cohorts.

Contamination. Twenty Ljubljana scans and the whole ISBI-2015 set are FLAMeS training data and were excluded. The remaining Ljubljana-10 come from a training site. The absence of overlap for the thick-slice cohorts rests on dataset descriptions, not on fingerprint forensics.

Confounding. Site, slice thickness, scanner, field strength, population (PediMS: children) and annotation protocol are confounded. The thickness bands describe these cohorts, not a controlled thickness effect. MS3SEG's protocol counts FLAMeS marks on "normal" periventricular hyperintensities as false.

The "1-mm" band. It mixes 10 natively thin scans with 110 resampled 2–5 mm acquisitions.

Sample sizes. PediMS has 28 scans of 9 children, and Ljubljana-10 has 10 patients. Bootstrap intervals with 9–10 clusters under-cover, and within-site P2 is impossible there.

Muslim. Its in-plane pixel size is estimated (±10 % in volume), which moves its size thresholds by one bin. It also spans about 20 uncontrolled centres with sequential annotation.

Geometry. 3D connected components on 4–8 mm slices can merge or split lesions. Candidate and expert-lesion rules are identical, but lesion counts are not comparable across grids.

Healthy controls. They come from one site and one dataset. The images were registered to a template (brains about 2.0 L), and their native acquisition and the controls' ages could not be verified. "Lesion-free" is an assumption about volunteers, not an expert read. The licence is non-commercial. Thirty of the 100 controls had been analysed with fold 0 before the addendum. The LST-AI arm on controls is pending. The DIS code used inherits a known nonconformity: it counts third-ventricle contact as periventricular. In patients this changed 0.08 % of lesion regions and no DIS decision, but it was not measured on controls.

Classification rule. "Holds" tolerates an upper bound up to nominal + 0.05 and means "no violation detected", not proof.

Repeated scans. P2, B2 and B3 calibrate on the first timepoint of each calibration patient, but the test sets include every timepoint of the test patients (MSLesSeg, PediMS). For follow-up scans, the exchangeability premise of these scan-level procedures is therefore approximate.

Pre-registration. The main protocol was not publicly time-stamped before its results (§5.1). A processing pilot preceded it.

Post-hoc analyses. Several explanatory analyses are post hoc. Two baselines and the training-site sensitivity analysis were added at the manuscript stage.

Not re-run. The per-scan region certificates of our earlier two-layer procedure were not re-run on the new sites, and Mondrian calibration by acquisition type was not tested.

Independent checks. The "independent" re-implementations were written by separate AI-agent sessions. No second human analyst has re-derived the numbers yet.

10 Conclusions

On five public MS cohorts and 100 healthy controls, conformal lesion-wise FDR control around a state-of-the-art segmenter showed no detected violation when moved to unseen sites and to thick 2D slices. Its calibration, however, did not transfer, and its per-scan counterpart failed on thick slices. The guarantee that survives is marginal: a property of the population of scans, not of the scan on the screen. It leaves little power and does not cover healthy brains. Fold ensembling buys power, not validity. Calibration must come from the deployed segmenter and the target acquisition. We translated these limits into a default-off software card that states what automatic lesion lists can and cannot be trusted for at a given slice thickness, and that withholds any false-lesion figure.

Declarations

Data availability. All data are public:

- MSLesSeg: Figshare, doi 10.6084/m9.figshare.27919209.
- Ljubljana 3D MR MS dataset: the laboratory's page and the open_ms_data mirror.
- MS3SEG: Figshare, doi 10.6084/m9.figshare.30393475.
- Muslim et al.: Mendeley Data, doi 10.17632/8bctsm8jz7.1.
- PediMS: Figshare, doi 10.6084/m9.figshare.28701065.
- SibBMS: https://ai.nsu.ru/files/sibbms/sibbms.zip, on request.

We do not redistribute images, masks, probability maps or per-candidate data. Some licences forbid derivatives (Ljubljana) or commercial use (Ljubljana, SibBMS). Aggregate result records (JSON) are in the code repository.

Code availability. Analysis code, tests, pre-registration documents, result records, the figure and table generators, and `NUMBERS_CHECK.md` are in the project repository under `research/paper_a/` and `docs/research/` (branch `research/paper-a-manuscript`). TODO (owner): confirm public visibility and the code licence, and archive a tagged release with a DOI before submission.

Ethics. This is a secondary analysis of publicly released, de-identified datasets used under their licences. No new data were acquired and no participant was contacted. The original studies report their own ethics approvals. TODO (owner): confirm with the author's institution whether a formal waiver is required.

Licences. See Table 2. Ljubljana is CC BY-NC-ND according to the laboratory's page; academic use is permitted, and we applied the stricter of the two stated licences. MSLesSeg data are CC BY 4.0. The software card (§7) shows detection figures whose ≤ 1.5 mm band includes the 10 Ljubljana scans. Using those figures in a commercial product requires clarifying that licence or re-deriving the band without Ljubljana; the card is off by default.

Funding. No external funding. FLAMeS and LST-AI inference ran on cloud CPU virtual machines paid from promotional research credit in a dedicated research project. All analyses in this paper ran on a local computer. TODO (owner): confirm.

Competing interests. N. G. Bonilla Vargas develops MSTool-AI, the MS imaging software in which the card of §7 was implemented. TODO (owner): state any commercial role, funding or patents.

Author contributions (CRediT). N. G. Bonilla Vargas: conceptualisation, methodology, software (with AI assistance, see below), formal analysis, writing (original draft and editing), project administration. TODO: co-authors. A statistician to review §3, §5 and §8.2, and a neuroradiologist to review §4, §6.9 and §7, are recommended before submission.

Use of AI assistance. The analysis code, the pre-registration documents, the result documents, the figures and this manuscript draft were produced with substantial assistance from an AI coding agent (Claude, Anthropic, used through Claude Code), working in the project repository under the author's direction. The two "independent re-implementations" (§5.7) were written in separate agent sessions. No AI system is an author. The author is responsible for the content. Every number is traced to a committed script and record (`NUMBERS_CHECK.md`).

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Appendix A — Additional tables

Table A1. Within-site calibration (D1), ensemble arm.

site	P1 α = 0.1: FDR · sens.	P1 α = 0.2: FDR · sens.	P2 γ = 0.1: FDX · sens.
MSLesSeg	0.016 · 0.102	0.046 · 0.309	0.087 · 0.042
Ljubljana-10	0.000 · 0.006	0.034 · 0.136	0.100 · 0.021
MS3SEG	0.041 · 0.062	0.075 · 0.198	0.100 · 0.040
Muslim	0.038 · 0.013	0.069 · 0.163	0.078 · 0.004
PediMS	0.024 · 0.070	0.066 · 0.154	abstains (8 calibration patients < 9)

Table A2. All procedures under leave-one-site-out calibration (D2), ensemble arm. Format as Table 5. The Ljubljana-10 column is a site seen in FLAMeS training.

procedure	α	MSLesSeg	Ljubljana-10	MS3SEG	Muslim	PediMS
P1 conformal BH (pre-registered, primary)	0.1	0.009 [0.000, 0.029] holds; sens. 0.010	0.000 [0.000, 0.000] holds; sens. 0.009	0.087 [0.059, 0.121] holds; sens. 0.201	0.082 [0.033, 0.137] holds; sens. 0.030	0.007 [0.000, 0.022] holds; sens. 0.064
P1 conformal BH (pre-registered, primary)	0.2	0.018 [0.004, 0.042] holds; sens. 0.089	0.011 [0.000, 0.033] holds; sens. 0.118	0.116 [0.093, 0.142] holds; sens. 0.381	0.100 [0.048, 0.163] holds; sens. 0.139	0.050 [0.016, 0.093] holds; sens. 0.164
one false candidate per calibration patient + BH [post hoc]	0.1	0.001 [0.000, 0.002] holds; sens. 0.007	0.000 [0.000, 0.000] holds; sens. 0.006	0.050 [0.033, 0.071] holds; sens. 0.117	0.025 [0.008, 0.046] holds; sens. 0.014	0.007 [0.001, 0.020] holds; sens. 0.036
one false candidate per calibration patient + BH [post hoc]	0.2	0.010 [0.004, 0.018] holds; sens. 0.072	0.011 [0.000, 0.033] holds; sens. 0.066	0.107 [0.084, 0.133] holds; sens. 0.336	0.071 [0.036, 0.113] holds; sens. 0.095	0.036 [0.017, 0.060] holds; sens. 0.131
hierarchical conformal e-values + e-BH [post hoc]	0.1	no selection (sens. 0)	no selection (sens. 0)	no selection (sens. 0)	no selection (sens. 0)	no selection (sens. 0)
hierarchical conformal e-values + e-BH [post hoc]	0.2	no selection (sens. 0)	0.119 [0.074, 0.169] holds; sens. 0.594	0.212 [0.188, 0.239] holds; sens. 0.559	no selection (sens. 0)	0.157 [0.115, 0.198] holds; sens. 0.589
B2 conformal risk control, per-scan FDP loss (pre-registered)	0.1	0.022 [0.000, 0.052] holds; sens. 0.004	0.000 [0.000, 0.000] holds; sens. 0.017	0.083 [0.040, 0.128] holds; sens. 0.092	0.157 [0.075, 0.245] inconclusive; sens. 0.028	0.000 [0.000, 0.000] holds; sens. 0.032
B2 conformal risk control, per-scan FDP loss (pre-registered)	0.2	0.042 [0.017, 0.073] holds; sens. 0.075	0.038 [0.000, 0.095] holds; sens. 0.100	0.094 [0.069, 0.125] holds; sens. 0.219	0.108 [0.070, 0.150] holds; sens. 0.201	0.061 [0.026, 0.120] holds; sens. 0.141
B3 Learn-then-Test, per-scan FDP loss, δ = 0.1 (pre-registered)	0.1	0.009 [0.000, 0.029] holds; sens. 0.000	0.000 [0.000, 0.000] holds; sens. 0.017	0.046 [0.018, 0.081] holds; sens. 0.066	0.134 [0.062, 0.226] inconclusive; sens. 0.018	0.000 [0.000, 0.000] holds; sens. 0.032
B3 Learn-then-Test, per-scan FDP loss, δ = 0.1 (pre-registered)	0.2	0.094 [0.073, 0.115] holds; sens. 0.460	0.072 [0.031, 0.116] holds; sens. 0.417	0.190 [0.161, 0.222] holds; sens. 0.507	0.151 [0.111, 0.188] holds; sens. 0.352	0.109 [0.065, 0.148] holds; sens. 0.487
B1 precision-matched threshold, no guarantee (pre-registered)	0.1	0.058 [0.043, 0.077] holds; sens. 0.336	0.046 [0.013, 0.082] holds; sens. 0.283	0.141 [0.117, 0.167] **violated**; sens. 0.440	0.120 [0.086, 0.158] inconclusive; sens. 0.285	0.097 [0.061, 0.137] holds; sens. 0.381
B1 precision-matched threshold, no guarantee (pre-registered)	0.2	0.210 [0.178, 0.241] holds; sens. 0.670	0.112 [0.066, 0.163] holds; sens. 0.579	0.222 [0.198, 0.247] holds; sens. 0.579	0.232 [0.183, 0.281] inconclusive; sens. 0.439	0.172 [0.125, 0.218] holds; sens. 0.620

Table A3. Primary cells with Ljubljana removed from every calibration pool [post hoc].

design · test site	cell	pre-registered pools (with Ljubljana-10)	Ljubljana removed from every pool [post hoc]
1-mm grid → MS3SEG	P1 α = 0.1	0.107 [0.075, 0.145] holds; sens. 0.258	0.098 [0.071, 0.131] holds; sens. 0.267
1-mm grid → MS3SEG	P1 α = 0.2	0.123 [0.099, 0.151] holds; sens. 0.404	0.124 [0.100, 0.151] holds; sens. 0.406
1-mm grid → MS3SEG	P2 γ = 0.1	0.21 [0.13, 0.30] **violated**; sens. 0.121	0.21 [0.13, 0.30] **violated**; sens. 0.121
1-mm grid → MS3SEG	P2 γ = 0.2	0.12 [0.06, 0.20] inconclusive; sens. 0.121	0.12 [0.06, 0.20] inconclusive; sens. 0.121
1-mm grid → Muslim	P1 α = 0.1	0.122 [0.062, 0.197] inconclusive; sens. 0.095	0.122 [0.062, 0.197] inconclusive; sens. 0.094
1-mm grid → Muslim	P1 α = 0.2	0.097 [0.058, 0.146] holds; sens. 0.235	0.097 [0.059, 0.147] holds; sens. 0.240
1-mm grid → Muslim	P2 γ = 0.1	0.29 [0.17, 0.44] **violated**; sens. 0.036	0.29 [0.17, 0.44] **violated**; sens. 0.036
1-mm grid → Muslim	P2 γ = 0.2	0.27 [0.16, 0.42] **violated**; sens. 0.036	0.27 [0.16, 0.42] **violated**; sens. 0.036
1-mm grid → PediMS	P1 α = 0.1	0.041 [0.009, 0.102] holds; sens. 0.120	0.044 [0.015, 0.089] holds; sens. 0.130
1-mm grid → PediMS	P1 α = 0.2	0.086 [0.056, 0.122] holds; sens. 0.326	0.088 [0.060, 0.122] holds; sens. 0.348
1-mm grid → PediMS	P2 γ = 0.1	0.07 [0.00, 0.24] inconclusive; sens. 0.053	0.07 [0.00, 0.24] inconclusive; sens. 0.053
1-mm grid → PediMS	P2 γ = 0.2	0.07 [0.00, 0.24] inconclusive; sens. 0.053	0.07 [0.00, 0.24] inconclusive; sens. 0.053
leave-one-site-out · MSLesSeg	P1 α = 0.1	0.009 [0.000, 0.029] holds; sens. 0.010	0.009 [0.000, 0.029] holds; sens. 0.010
leave-one-site-out · MSLesSeg	P1 α = 0.2	0.018 [0.004, 0.042] holds; sens. 0.089	0.018 [0.004, 0.042] holds; sens. 0.085
leave-one-site-out · MSLesSeg	P2 γ = 0.1	0.02 [0.00, 0.06] holds; sens. 0.001	0.02 [0.00, 0.06] holds; sens. 0.000
leave-one-site-out · MSLesSeg	P2 γ = 0.2	0.03 [0.00, 0.07] holds; sens. 0.004	0.02 [0.00, 0.06] holds; sens. 0.001
leave-one-site-out · MS3SEG	P1 α = 0.1	0.087 [0.059, 0.121] holds; sens. 0.201	0.087 [0.059, 0.121] holds; sens. 0.202
leave-one-site-out · MS3SEG	P1 α = 0.2	0.116 [0.093, 0.142] holds; sens. 0.381	0.115 [0.093, 0.142] holds; sens. 0.382
leave-one-site-out · MS3SEG	P2 γ = 0.1	0.16 [0.09, 0.25] inconclusive; sens. 0.087	0.14 [0.08, 0.22] inconclusive; sens. 0.082
leave-one-site-out · MS3SEG	P2 γ = 0.2	0.12 [0.06, 0.20] inconclusive; sens. 0.087	0.10 [0.05, 0.18] inconclusive; sens. 0.082
leave-one-site-out · Muslim	P1 α = 0.1	0.082 [0.033, 0.137] holds; sens. 0.030	0.082 [0.033, 0.137] holds; sens. 0.030
leave-one-site-out · Muslim	P1 α = 0.2	0.100 [0.048, 0.163] holds; sens. 0.139	0.102 [0.049, 0.165] holds; sens. 0.137
leave-one-site-out · Muslim	P2 γ = 0.1	0.18 [0.08, 0.31] inconclusive; sens. 0.015	0.16 [0.06, 0.29] inconclusive; sens. 0.012
leave-one-site-out · Muslim	P2 γ = 0.2	0.18 [0.08, 0.31] inconclusive; sens. 0.015	0.18 [0.08, 0.31] inconclusive; sens. 0.015
leave-one-site-out · PediMS	P1 α = 0.1	0.007 [0.000, 0.022] holds; sens. 0.064	0.007 [0.000, 0.022] holds; sens. 0.063
leave-one-site-out · PediMS	P1 α = 0.2	0.050 [0.016, 0.093] holds; sens. 0.164	0.050 [0.016, 0.093] holds; sens. 0.163
leave-one-site-out · PediMS	P2 γ = 0.1	0.00 [0.00, 0.12] holds; sens. 0.025	0.00 [0.00, 0.12] holds; sens. 0.025
leave-one-site-out · PediMS	P2 γ = 0.2	0.00 [0.00, 0.12] holds; sens. 0.025	0.00 [0.00, 0.12] holds; sens. 0.025

Appendix B — Claims of our earlier drafts that this audit retracts or changes

earlier claim (draft)	status now	where
Evaluation on all 30 Ljubljana scans as external (v0.1)	**retracted**: 20 are FLAMeS training data; only 10 are used, and their site is flagged as seen	§4.2
"0 of 200 healthy scans certified as DIS, 95 % CI ≤ 1.8 %" (v0.1)	**retracted**: 20 resplits of 30 scans; honest ceiling about 10 % (0/30, one-sided bound 9.5 %);	§6.9

	replaced by 100 distinct controls: 19/100 meet the DIS rule without selection, 0–3/100 after selection	
"0/30 healthy controls certified as DIS" (v0.2, two-layer certificates, fold 0)	**superseded**, not re-run	§6.9, §9
ISBI-2015 segmenter-shift result (v0.2)	**withdrawn**: ISBI-2015 is FLAMeS training data	§4.2
"Lesion-level AUROC 0.70–0.75; the score is the bottleneck" (v0.2)	**changed**: true for a single fold; the 5-fold ensemble scores 0.78–0.82	§6.1
"A segmenter change collapses power — the safe direction" (v0.2)	**changed**: the reverse direction violates the guarantee on the same images (expected from score scales)	§6.10
"The guarantee held in every held-out cell" (first v0.3 text)	**changed**: no violation detected; one site undecided; three abstention cells; two small sites	§6.2
"Validity and power change with slice thickness" (first v0.3 text)	**changed**: confounded with site; small within sites	§6.6
"The ensemble restores validity" (first v0.3 text)	**changed**: power gain; raises FDR on one site; two new violated cells	§6.7
"Ten local patients restore the per-scan guarantee" (first v0.3 text)	**qualified**: by near-abstention (0.65–3.85 % of lesions)	§6.8
"Conformal BH is the only procedure without a violated 1-mm → thick cell" (v0.3 results)	**corrected**: Learn-then-Test had none either	§6.4
"MSLesSeg native geometry unpublished" (first v0.3 text)	**corrected**: native 2–5 mm per the dataset's acquisition table	§4.1
Earlier claims of v0.1 corrected in v0.2 ("7–13× the floor", "conformal breaks across MS site shift", a transport p-value, candidate AUROC without a source)	remain withdrawn	—

Appendix C — Reproducibility map

result	**script**	**record**
pre-registered protocol and deviations	—	`docs/research/paper/PROTOCOL-v0.3.md` (commit 9951494 and later entries)
input hashes, grids, fold-0 reproduction	`research/paper_a/v03_inventory.py`	`docs/research/records/2026-10-21_paper_a_v03_inventory.json`
candidates, labels, gate G1	`research/paper_a/v03_candidates.py`	`docs/research/records/2026-10-21_paper_a_v03_candidates.json`
Tables 1, 4–9, A1, A2 (pre-registered parts)	`research/paper_a/v03_analysis.py` (+ `v03_metrics.py`, `v03_data.py`, `baselines.py`)	`docs/research/records/2026-10-21_paper_a_v03_results.json`
post-hoc explanations (§6.2, §6.3, §6.6, §6.7)	`research/paper_a/v03_posthoc.py`	`docs/research/records/2026-10-21_paper_a_v03_posthoc.json`
MSLesSeg native thickness [post hoc]	`research/paper_a/v03_mslesseg_native.py`	`docs/research/records/2026-10-21_paper_a_v03_mslesseg_native.json`
SUB, LR-hier, Ljubljana-out [post hoc] (Table 5, A2, A3)	`research/paper_a/v03m_manuscript_posthoc.py`	`docs/research/records/2026-10-31_paper_a_manuscript_posthoc.json`
controls: protocol	—	`docs/research/paper/PROTOCOL-v0.3-addendum-controls.md` (commit caa47e7)
controls: inventory, contamination check, frozen calibration	`research/paper_a/v03c_inventory.py`, `v03c_calibration.py`	`docs/research/records/2026-10-24_paper_a_v03c_{inventory,calibration}.json`
Table 10, §6.9	`research/paper_a/v03c_controls.py`	`docs/research/records/2026-10-24_paper_a_v03c_controls.json`
independent control counts	`research/paper_a/v03c_controls_check.py`	`docs/research/records/2026-10-24_paper_a_v03c_controls_check.json`
figures	`docs/research/paper/manuscript/figures/make_figures.py`	`figures/figures_manifest.json`
tables in this manuscript	`docs/research/paper/manuscript/tools/make_tables.py`	(generated from the records above)
number-by-number trace	`docs/research/paper/manuscript/tools/check_numbers.py`	`docs/research/paper/manuscript/NUMBERS_CHECK.md`

Appendix D — Open items before submission (owner checklist)

1. Co-authors: a mathematical statistician and a neuroradiologist (contributions, review of §3, §5, §8.2, §4, §6.9 and §7).
2. A human re-derivation of the primary numbers, and archiving of the patient-analysis re-implementation code, which is not in the repository.
3. Refresh the novelty search (the 2026-10-17 search ran partly without web-search quota).
4. Check the bibliography entries marked OWNER-CHECK in `references.bib` (Jin & Candès volume; Barber et al. pages; HD-BET; MSLesSeg title).
5. Licences: the Ljubljana CC BY-NC-ND question for the software card; SibBMS (two licences stated).
6. Affiliation, ORCID, e-mail, funding, competing interests, ethics waiver, and code visibility, licence and DOI.
7. Optional before submission: the LST-AI control arm; a second-site, natively acquired, expert-read control set; a prospective test of Mondrian calibration by acquisition type, pre-registered in a public registry.